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Attention Restoration Theory II: a systematic review to clarify attention processes affected by exposure to natural environments

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ABSTRACT
Attention Restoration Theory (ART) predicts exposure to natural environments may lead to improved cognitive performance through restoration of a limited cognitive resource, directed attention. A recent review by Ohly and colleagues (2016) uncovered substantial ambiguity surrounding details of directed attention and how cognitive restoration was tested. Therefore, an updated systematic review was conducted to identify relevant cognitive domains from which to describe elements of directed attention sensitive to the restoration effect. Forty-two articles that tested natural environments or stimuli against a suitable control, and included an objective measure of cognitive performance, had been published since July 2013. Articles were subjected to screening procedures and quality appraisal. Random effects meta-analyses were performed to calculate pooled effect sizes across 8 cognitive domains using data from 49 individual outcome measures. Results showed that working memory, cognitive flexibility, and to a less-reliable degree, attentional control, are improved after exposure to natural environments, with low to moderate effect sizes. Moderator analyses revealed that actual exposures to real environments may enhance the restoration effect within these three domains, relative to virtual exposures; however, this may also be due to differences in the typical lengths of exposure. The effect of a participants' restoration potential, based upon diagnosis or fatigue-induction, was less clear. A new framework is presented to qualify the involvement of directed attention-related processes, using examples of tasks from the three cognitive domains found to be sensitive to the restoration effect. The review clarifies the description of cognitive processes sensitive to natural environments, using current evidence, while exploring aspects of protocol that appear influential to the strength of the restoration effect.

KEYWORDS
Executive functions; greenspace; mental fatigue; restorative environments

Introduction

From a flashing advertisement that reminds us we are hungry, to a beeping phone that keeps us updated; modern urban environments contain distractors that are designed to be distracting. To navigate such environments, the brain must rely heavily on cognitive resources that direct attention towards a desired goal while simultaneously ignoring distraction. However, the capacity to direct attention in such a way cannot be sustained indefinitely, which contributes to the waxing and waning of our ability to concentrate throughout the day. Prolonged use of this cognitive resource leads to a state of mental fatigue that needs to be overcome. The increasing cognitive demand of modern environments has rapidly expanded counter-movements that promote regular use of natural environments to maintain a healthy lifestyle. The expansion of nature-based programs in education (Barfod et al. 2016) and the suggestion that health-care providers should be offering prescriptions for exposure to natural environments as a viable intervention for a range of conditions (Zarr, Cottrell, and Merrill 2017), created an urgency to understand the mechanisms underlying the often reported positive effects on brain activity.

In 2016, Ohly and colleagues conducted a systematic review and meta-analysis of the objective evidence available on the effects natural environments have on cognitive processes, following the leading theory in the area, Attention Restoration

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Supplemental data for this article can be accessed here.

Table 1. Characteristic of studies identified by systematic literature search, ordered by: randomized crossover trials, randomized controlled trials, quasi-experimental studies, and natural experiments.

Author, Year, (sub-study)	Study design	Country, setting	*n*	Study characteristics: population; gender; age	Intervention and control characteristics: exposure type; activity; subgroup/environments; duration	Cognitive performance measure
Geniole et al. (2016)	RCT (w)	Canada; naturalised landfill, urban area	31	University students; male only; m = 24.61 years	Actual; walking; naturalised landfill vs. business/ commercial urban area (some natural stimuli); 15′00	Stroop Task
Jung et al. (2017)	RTC (w)	United States; laboratory	40	Heart failure patients, healthy matched adults; 47.5% female; m = 59.2 years	Virtual; viewing images; nature scenery vs. urban views; 07′00	Multi-Source Interference Task; Digit Span Forwards; Digit Span Backwards; Trail Making Test A, B; StroopTest
Rider and Bodner (2016) (1)	RCT (w)	Canada; university campus	24	University students; 83.3% female; m = 22 years	Actual; walking; natural vs. urban vs. indoor; 10′00	Free Recall Memory (words); Forced-Choice Recognition Memory
Rider and Bodner (2016) (2)	RCT (w)	Canada; university campus	24	University students; 79.2% female; m = 22 years	Actual; walking; natural vs. urban vs. indoor; 10′00	Free Recall Memory (words); Forced-Choice Recognition Memory
Rogerson et al. (2016)	RCT (w)	United Kingdom; universty campus	24	General population (n = 13), university staff (n = 1), university students (n = 10); 79.2% female; m = 35.1 years	Actual; cycling; view of sportsfield, tress vs. indoor view; 15′00	Digit Span Backwards
Sahlin et al. (2016)	RCT (w)	Sweden; urban areas, nature reserve	51	General population; 76.5% female; m = 45 years	Actual; guided relaxation; outdoor natural vs. Indoor; 30′00	Necker Cube Pattern Control
Schutte, Torquati, and Beattie (2017)	RCT (w)	United States; urban areas	67	Children (four age groups); 52.2% female; m = 4.53, m = 5.48, m = 7.4 years, m = 8.5 years	Actual; walking; natural vs. urban; 20′00	Spatial Working Memory Task; Go/NoGo Task; Continuous Performance Task; Digit Span Backwards
Sonntag-Öström et al., (2014)	RCT (w)	Sweden; city and forest areas	20	Patients with Exhaustion Disorder; female only; m = 41.6 years	Actual; walking, relaxing; forest by lake vs. rock outcrop, vs. spruce forest vs. city; 10′00, 40′00	Necker Cube Pattern Control
Triguero-Mas et al. (2017)	RCT (w)	Spain; Collserola NP, Castelldefels Beach, Eixample (neighbourhood)	26	General population adults, lowest 50th percentile scores on MHI-5; 57.69% female; median = 44.32 years	Actual; national park, beach vs. neighbourhood environment; 30′00, 180′00	Digit Span Backwards
Berry et al. (2014)	RCT (b)	United States; laboratory	185	University students; 58% female; m = 20.88 years	Virtual; viewing images; natural vs. urban vs. geometric; EDTP	Delayed Discounting Task
Berry et al. (2015)	RCT (b)	United States; laboratory	43	University students; 60.5% female; m = 22.5 years	Virtual; viewing images; natural vs. urban; EDTP	Delayed Discounting Task; Interval Bisection Task
Beute and De Kort (2014), (1)	RCT (b)	The Netherlands; laboratory	50a	University students; 43.3% female; m = 22.2 yearsa	Virtual; viewing images; natural vs. urban; 03′00	Stroop Task; Operation Span Task;
Beute and De Kort (2014), (2)	RCT (b)	The Netherlands; laboratory	61b	University students; 42.1% female; m = 21.1 yearsb	Virtual; viewing images; natural vs. urban; 03′00	2-Back Task
Bratman et al. (2015)	RCT (b)	United States; university surrounds	60	General population; 55% female; m = 22.9 years	Actual; walking, taking photos; urban park vs. busy street; 50′00	Operation Span Task; Change Detection Task; Attention Network Task; Digit Span Backwards

(*Continued*)

Table 1. (Continued).

Author, Year, (sub-study)	Study design	Country, setting	*n*	Study characteristics: population; gender; age	Intervention and control characteristics: exposure type; activity; subgroup/environments; duration	Cognitive performance measure
Chaw and Lau (2015), (1)	RCT (b)	Hong Kong, laboratory	42	University students; 66.7% female; m = 20.81 years	Virtual; viewing images, imagination; natural vs. non-viewing (resting); 06′00	Unsolvable Anagrams
Chaw and Lau (2015), (2)	RCT (b)	Hong Kong; laboratory	58	University students; 62.1% female; m = 19.79 years	Virtual; viewing images, imagination; natural vs. urban; 04′00	Logical Reasoning - Graduate Exam Record
Chaw and Lau (2015) (3)	RCT (b)	United States; laboratory	185b	General population; 66% female; m = 38.58 yearsb	Virtual; viewing images, clicking on attractive areas; natural vs. urban; 01′30	Unsolvable Anagrams
Craig et al. (2015)	RCT (b)	United States; laboratory	48	University studentsc; 45.8% female; m = 20.1 years	Virtual; viewing images; natural vs. urban; 04′10	Sustained Attention to Response Task
Emfield and Neider (2014)	RCT (b)	United States; laboratory	192	University students; 66.1% female; m = 19.8 years	Virtual; viewing images/listening to sounds/both; natural vs. Urban; 07′00 - 10′00	Digit Span Backwards; Attention Network Task; Functional Field of View
Evensen et al. (2015)	RTC (b)	Norway; laboratory	85	University students; 67.1% female; m = 24.9 years	Actual; viewing; window + plants vs. window + inanimate objects vs. window only vs. no window + plants vs. no window + inaminate objects vs. No window only; EDTP	Reading Span Task
Gamble, Howard, and Howard (2014)	RCT (b)	United States; laboratory	56	University students, general population; NR; m = 20.54 years, m = 69.1 years	Virtual; viewing images; natural vs. urban; 05′50	Attention Network Task; Digit Span Backwards
Greenwood and Gatersleben (2016)	RTC (b)	United Kingdom; high school, South West London	120	High school students; 55% female; 16–18 years	Actual; relaxing vs. talking with peer vs. Playing game on mobile phone; Natural outdoor vs. small indoor room; 20′00	Necker Cube Pattern Control*
Haga et al. (2016)	RTC (b)	Sweden; university campus	90	University students; 68% female; m = 24.76 years	Virtual; pink noise stimuli - attributed to natural (waterfall) or non-natural (machinery) source; 03′00	Attention Network Task
Han (2017)	RCT (b)	Taiwan; university campus	116	University students; 55.2% female; m = 20.85 years	Actual; walking - ‘high natural’ vs. ‘low natural’; jogging - ‘high natural’ vs. ‘low natural’; 15′00	Forward Spatial Span, Digit Span Backwards
Jenkin et al. (2017)	RCT (b)	England;	75	Children; 50.6% female; 8–11 years	Virtual; viewing video of natural scenes vs. Viewing video of urban scenes; 03′00	Delay of Gratification; Stroop
Johansson et al. (2015)	RTC (b)	Sweden; Gothenburg Botanical Gardens	34	Aquired brain injury patients, reporting mental fatigue; 82.3% female; m = 48.0, 46.6, 51.2 years	Actual; walking; Nature walks in groups vs. face-to-face MBSRd vs. internet MBSRd; 90′00/week (x 8)	Digit Symbol Coding; Attentional Blink Task
Lee et al. (2015)	RCT (b)	Australia; laboratory	150	University students; 71% female; m = 20 years	Virtual; Viewing city with green roof vs. Viewing city without green roof; 00′40	Sustained Attention to Response Task
Li and Sullivan (2016)	RCT (b)	United States; high schools	94	High school students; 56.4% female; NR	Actual; Greenspace window view vs. builtspace window view vs. no window view; EDTP (10′00 rest before post test)	Digit Span Fowards + Digit Span Backwards (summary score)
Lin et al. (2014)	RCT (b)	Taiwan; laboratory (classroom)	138	University students; 52.9% female; NR	Virtual; viewing images; urban street with simulated natural stimuli (minimal, moderate, or heightend awareness towards natural stimuli) vs. urban street without natural stimuli; 01′40	Digit Span Backwards
Lobo, Ramachandran, and Tiwari (2015)	RCT (b)	India; laboratory	40	Students; 50% female; m = 22.95 years	Virtual; viewing images + imagination; natural (pleasant, neutral) vs. urban (unpleasant vs. neutral); 00′50	Navon Global-Local Task

(*Continued*)

Table 1. (Continued).

Author, Year, (sub-study)	Study design	Country, setting	*n*	Study characteristics: population; gender; age	Intervention and control characteristics: exposure type; activity; subgroup/environments; duration	Cognitive performance measure
Lymeus, Lundgren, and Hartig (2017)	RCT (b)	Sweden; small classroom	51	University students; 72.5% female; m = 25 years	Virtual; mindfulness meditation with nature images vs. Conventional mindfulness meditation vs. Rest with nature images; 15′00	Letter-Digit Substitution Test
Pilotti et al. (2014)	RCT (b)	United States; laboratory	63	University student advisors; 61.9% female; m = 31.79 years	Virtual; viewing video; natural vs. urban; 15′00	Auditory Oddball Task
Studente, Seppala, and Sadowska (2016)	RCT (b)	United Kingdom; laboratory	108	Universty students; NR; NR	Actual; viewing; window + plants vs. no window + no plants + green paper vs. No window + no plants; EDTP	Alternative Uses Test; 30 Circles Test
Tanaka et al. (2013)	RCT (b)	Japan; housing	16	General population; 100% female; median = 43.5 years	Actual; viewing; nature vs. no nature; 30′00	Advanced Trail Making (A, B, C)
Van Rompay and Jol (2016)	RCT (b)	The Netherlands; laboratory	120	High school students; 50% female; m = 13.8 years	Virtual; viewing images + imagination; natural vs. urban; EDTP	Tests for Creative Thinking - Drawing Production
Wang et al. (2016)	RCT (b)	China; laboratory	140	University students; 50% female; m = 22.38 years	Virtual; viewing videos; natural vs. urban; 08′00	Digit Span Backwards
Wang et al. (2017)	RCT (b)	China; laboratory	118	University students; 78.8% female; m = 21.23 years	Virtual; viewing videos; natural vs. urban; 10′00	Taylor's Aggression Paradigm (Reaction Time Competition)
Zhang, Kang, and Kang (2017)	RCT (b)	China; urban park	70	University students; 51.4% female; m =	Actual; resting; urban park enivronment (within-subjects) with primarily bird and insect sounds vs. Primarily periodic machinery sound vs. Primarily continous transport sound; 40′00	Complement Numbers Test (simple addition fluency)
Ferraro (2015)	Quasi-experiment	United States; Canoe trip in Minnesota	25	University students; NR; m = 18.71, 18.38 years	Actual; hiking; wilderness hike vs. indoor control; six days	Remote Associations Test
Kelz, Evans, and Röderer (2015)	Quasi-experiment	Austria; high schools	133	High school students; 48.5% female; 13–15 years	Actual; Green school yard design vs. control school yard n; 6–7 weeks	Attention Network Task
Van den Berg et al. (2016)	Quasi-experiment	The Netherlands; primary schools	170	School children; 42.9% female; m = 9 years	Actual; viewing; "green wall" with living plants vs. no green wall; 4 months	Digit Letter Substitution Test; Sky Search Task
Dadvand et al. (2015)	Natural experiment	Spain; school and residential areas in Barcelona	2623	School children; 50% female; m = 8.5 years	Actual; Exposure to greenspace (residential, commute to school, school); Cognitive testing progress over 12 months	n-Back Test; Attention Network Task
Dadvand et al. (2016)	Natural experiment	Spain; residential areas in Sabadell and Valencia	888; 978	Children; 48.8% female (K-CPT & ANT); birth-7 years	Actual; Residential exposure to greenspace; K-CPT at 4–5 years, ANT at 7 years	Conner's Kiddie Continuous Performance Test; Attention Network Test
Ulset et al. (2017)	Natural experiment	Norway; daycare centres	562	Young children; 53% female; m = 52.45 months	Actual; time spent outdoors during daycare hours; 4 years	Digit-Span Forwards Test
Ward et al. (2016)	Natural experiment	New Zealand; intermediate (middle) schools	72	Intermediate (middle) school students; 56.9% female; m = 12.66	Actual; Exposure to greenspace measured using portable GPS; 7 days	CNS Vital Signs Neurocognitive Test Battery
Zijlema et al. (2017)	Natural experiment	Spain, The Netherlands, United Kingdom	1493-1602m	General population adults; 54,1% female; m = 48 years	Actual; Residential distance to greenspace, residential surrounding greenness; n/a	Color Trails Test

Notes: RCT = randomized controlled trial; w = within-participants; b = between-participants; MHI = Mental Health Index; EDTP = exposure during task performance; MBSR = mindfulness-based stress reduction; NR = not reported; K-CPT = Kiddies Continuous Performance Task; ANT = Attention Network Task.

Theory (ART). According to ART (Kaplan 1995; Kaplan and Kaplan 1989), the mental fatigue that is associated with a depleted capacity to direct attention may be overcome by spending time in environments rich in natural stimuli. A central premise of ART suggests that natural stimuli are intrinsically fascinating in a way that evokes a type of attention that does not require effortful fixation or cognitive effort (Kaplan 1995). As the brain engages externally driven and effortless attention, the capacity to direct attention in an effortful manner is restored. The process of restoration is not necessarily exclusive to natural environments; however, these environments are nominated as being particularly suitable for covering the 4 characteristics of a restorative environment (Kaplan 1995). According to the theory, environments that contain so-called *softly fascinating* stimuli should be optimally restorative when they also evoke a feeling of *being away* or separated from cognitively fatiguing situations; have a sense of *extent* which can be literal in the sense of physical size or conceptual by evoking the feeling of being connected to something larger; and be *compatible* with the individual's desires and goals so that it does not impede but supports the realization of these desires (Kaplan 1995).

Ambiguity surrounding directed attention in ART

The phenomenon of attention restoration is believed to be captured by studies reporting improvements on cognitive tasks after exposure to a natural environment or natural stimuli, relative to a suitable control. Ohly and colleagues' (2016) systematic review was timely in pointing out the ambiguity in the field surrounding the construct of directed attention, reflected in the diversity of tasks used to measure the effect. The Digit Span Forwards (DSF) and Digit Span Backwards (DSB) Tasks were two of the three outcome measures that showed significant positive effects in their meta-analysis that pooled data across the literature base. These tasks are traditionally considered working memory tasks with varied loads of cognitive effort or demands on other processes, such as attention (Conway et al. 2005). The remaining task found to be significantly affected by natural environments was the Trail Making Test B, which is generally considered a measure of cognitive flexibility (Kortte, Horner, and Windham 2002) or attentional control that allows switching between two rule sets (Arbuthnott and Frank 2000). Thus, tasks in which a restoration effect is most reliably detected seem to rely on directed attention capacity in different ways and to varying degrees. In this sense, the exact aspects of directed attention which is influenced by natural environments remain obscured. One task that isolates the ability to direct attention while simultaneously ignoring distraction is the Attention Network Task (Fan et al. 2005). Unfortunately, when Ohly and colleagues (2016) performed their literature search in 2013, there had been only one study using this refined measure of directed attention that isolates the ability from other cognitive processes. Therefore, there is not currently sufficient evidence to determine whether natural environments only restore directed attention indirectly through affecting additional related cognitive processes activated during task performance or whether restoration is also possible when the ability to direct attention is isolated by a task's characteristics.

Ohly and colleagues (2016) concluded that the field would benefit from identifying measures of directed attention that are most likely to be sensitive to natural environments and by then considering the specific characteristics of the tasks and associated attentional processes that appear to be driving the effect. They argued that there should be a goal to reach an agreement on what should constitute a measure of directed attention and to establish an optimal testing protocol, so that the restoration effect can be tested consistently across multiple studies, leading to a more reliable synthesis of data in the future.

Challenges faced by Ohly and colleagues (2016)

In a letter to the editor in response to the review, Hartig and Jahncke (2017) echoed the concerns raised by the article, calling for a tightening between theory and measurement of effects. Their commentary was positive and praised the substantial effort and contribution by the authors. They did, however, also indicate additional concerns over the meta-analysis itself and warned against putting too much emphasis on such publications without considering the finer details. The problems arising from Ohly and colleagues (2016) meta-analysis can be attributed to the diversity of methods they uncovered, which may have influenced their conclusions.

Firstly, the need to build links between protocol and theory is exemplified by the inclusion or exclusion of a cognitive loading task in attention restoration studies. A cognitive loading task is sometimes included prior to baseline attention measures or environmental treatment to induce a state a mental fatigue. Because the effect of a natural environment rests on the premise that an individual experiences restoration from a fatigued state, the inclusion of such a task seems critical to the existence or at least the strength of the effect being studied. ART itself predicts that participants who are cognitively fatigued prior to an environmental treatment will have a higher restoration potential and are therefore more likely to demonstrate greater improvement in task performance than those who are not. Thus, on a theoretical level, combining effect sizes using fatigued and non-fatigued participants, as Ohly et al. (2016) did, seems problematic. Incidentally, one of Hartig's own studies (Hartig et al. 2003) in which fatigue induction was used as a between-subjects factor, showed no marked effect of the fatigue-inducing task. Despite this result going against the predictions of ART, it seems important to consider the influence of fatigue induction when calculating effect sizes at a meta-analytic level.

Secondly, Ohly and colleagues (2016) were faced with the difficulty of baseline imbalances and studies that did not include baseline measures from which to measure changes in cognitive performance related to the environmental treatments. They noted that when baseline measures were included, the vast majority of authors in the field preferred to analyze change scores (post-test minus pre-test) to measure treatment effect, rather than using ANCOVA on post-test scores where the baseline is included as a covariate, as is preferred for randomized treatment studies (Van Breukelen 2013). The calculation of effect sizes related to change scores is challenging for meta-analysis, due to the difficulty of obtaining the standard deviations (SD) of the standardized mean difference. To calculate effect sizes based on change scores, Ohly et al. (2016) would have had to obtain the Pearson's r correlation between the pre-test scores for each study, which is seldom calculated or reported. Imputation of r for calculating SD in this manner is possible, though not recommended (Cochrane Collaboration 2008). Therefore, Ohly and et al (2016) conducted their meta-analysis using only post-test attention scores following the environmental interventions. Even with randomized within-subject designs, the baseline is necessary to control for intra-individual differences that occur between sessions. In attention restoration research, many studies use separate days for each environmental treatment. Therefore, without baseline measurements, it is unclear whether post-test measures are related to the treatment of the session or an unforeseen spurious variable that differed between the testing days, such as amount or quality of sleep the night before.

Clarifying directed attention and exploring moderator variables

An updated systematic literature review and meta-analysis was conducted to supplement the investigation by Ohly and colleagues (2016). The primary aim was to expand the current understanding of directed attention within the field to remove ambiguity surrounding the cognitive processes sensitive to the restoration effect associated with exposure to natural environments. The secondary aim was to explore the influence of protocol characteristics, through moderator analysis, to begin a dialogue towards optimal protocols for attention restoration research such that greater homogeneity might be achieved across future studies.

To address these aims, a subgroup of studies was selected from the original (Ohly et al. 2016) and updated literature searches where baseline measures prior to environmental exposure had been included. The following were recorded: 1) timing of baseline measurement (Figure 1); 2) the average baseline score; 3) whether restoration potential of participants was heightened, either

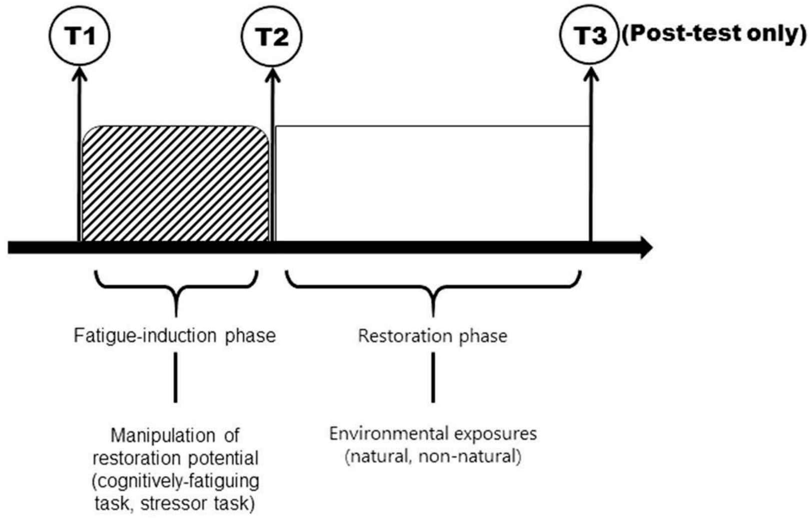

Figure 1. Timeline of a typical randomized controlled trial in attention restoration research. T1 and T2 represent commonly used baseline measurements, occurring before or after the fatigue induction phase. T3 may represent the timing of post-test or single measurement of cognitive performance when no baseline measurement is taken.

by an experimental fatigue-induction phase or through participant characteristics; and 4) type of environmental exposure (virtual stimuli presented in highly controlled laboratory settings or actual exposures taking place in real environments).

For initial analysis, outcome measures were grouped based on relevant cognitive domains, in order to provide pooled mean effect sizes based on how directed attention is recruited. This enabled us to discuss the findings of our analyses in relation to underlying attentional processes by considering the specific characteristics of the cognitive domains and outcome measures shown to be most sensitive to natural environments. For moderator analysis, the extent to which the pooled mean effect estimates for each cognitive domain could be explained by a balance of baseline scores, restoration potential of participants, and type of environmental exposure was explored. The results are discussed by presenting a three-axis framework that may be useful in differentiating and qualifying the role of directed attention in each task or cognitive domain. The proposed framework may be used to guide theory development and future research, particularly in terms of task selection.

Methods

The search, collection, and review of literature was conducted following the guidelines put forth by the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement, which aimed to optimize systematic reviewing of scientific research (Moher et al. 2009). Details of the search protocol were published on the PROSPERO database (Reference: CRD42017057718).

Literature search

A systematic search strategy designed to review the objective evidence surrounding ART had been developed independently, prior to the publication of Ohly et al. (2016) meta-analysis, with a high level of similarity. The original search strategy was created to capture existing research that contained points of interest within all three key concept domains. Firstly, all research that investigated cognition after exposure to the natural environment or natural stimuli was included. The term ‘natural environment’ is broad and in psychological research may be simply used to describe a non-laboratory setting. Therefore, many relevant search terms and phrases that were inspired by preliminary searches were included. Large variation was found in the way authors chose to describe the use of natural environments in their research. For example, Taylor, Kuo, and Sullivan (2002) used 6 different ways in their abstract to express the concept of nature (i.e., “views of nature,” “contact with nature,” “near-home nature,” “nearby nature," "naturalness," and "green space"). Our definition of *exposure* to the natural environment was broad and therefore included anything from passively viewing nature scenes on a computer screen to active activities (e.g., running) in real environments. Secondly, only studies that used measures of cognitive function were considered. Therefore, search terms related to cognitive abilities and attention restoration were included. Finally, only studies that measured cognitive ability using objective standardized cognitive tasks were included. Thus, a final key concept related to tasks, along with additional relevant terms was included in the search strategy.

Based upon the almost complete overlap of search terms and studies uncovered in our original search strategy and those reported by Ohly et al. (2016), we were confident in using our search string to follow-up their work. However, to address discrepancies, two search terms ("garden*" and "attention restorat*") used by Ohly and colleagues (2016) were added that were not included in our original search string. The following search string was applied to systematically screen titles, abstracts, and keywords from 6 databases (PsycINFO, SPORTDiscus, Medline, Embase, Scopus, and Web of Science) in order to cover a broad range of scientific disciplines:

("natural environment" OR "natural area*" OR "natural space*" OR "green area*" OR "green space*" OR greenspace* OR forest OR woodland* OR "nature exposure*" OR "exposure* to nature" OR "nature experience*" OR "experience* in nature" OR "contact with nature" OR "nature contact" OR "interact* with nature" OR "restorative environment*" OR "natur* view*" OR garden* OR "attention restorat*")*

AND

(cognit OR attention* OR memory OR concentrat* OR "executive function*" OR "problem solving" OR inhibition OR impulsiv* OR "mental fatigue" OR self-regulation")*

AND

(test OR task OR battery OR exam*)*

Study selection and eligibility criteria

The final systematic search of databases took place during November 2017. A number of eligibility criteria were put in place in order to capture literature relevant to our purpose. Limits were set during the literature search in order to only include peer-reviewed original research articles published after July 2013, the month in which Ohly et al. (2016) conducted their search. The articles were screened by title and abstract in order to identify potentially relevant studies. Full-text screening took place for all articles that appeared relevant after the first two screenings. Full texts were excluded from the final review if they fell under any of the exclusion criteria that were defined during screening. Specifically, studies were excluded that (a) were not experimental studies; (b) did not use a natural environment or natural stimuli; (c) did not include a suitable control group or comparison environment (e.g., natural vs urban); or (d) did not include objective outcome measures derived from standardized cognitive tasks. The search was not limited based on participant characteristics such as gender, age, or diagnostic status of participants. Nor was the search limited by country, culture, or presence of water in the natural environment. Research published in languages other than English was not included.

Data extraction

The same study characteristics were selected as presented by Ohly et al. (2016) review. Specifically, study design, participant details, setting details for natural and control conditions, type of exposure (virtual, physical), engagement with the environment, duration of exposure, and cognitive measures were recorded. Data based on pre- and post-test means and standard deviations were extracted. Supplementary information was added regarding the restoration potential of participants. This included details of any reported fatigue induction using a stressor task or cognitive loading task, the duration of the induction phase, and whether participant characteristics might contribute to restoration potential, for example, by having a relevant diagnosis.

Quality appraisal

To further the continuity with Ohly et al. (2016) investigation, the same quality appraisal system was used to rate the included studies. Their quality appraisal scale contained a combination of indicators that were derived from three rating systems previously developed to identify potential bias and quantify the robustness of evidence. For specific descriptions of each individual indicator, see Table 2 in Ohly and colleagues (2016). A two-factor checklist was added to the quality appraisal system to identify studies that were suitable for inclusion in our meta-analysis.

It is understood that not all information could be provided in published manuscripts, so in agreement with Ohly et al. (2016), authors were directly contacted for clarifications when an initial assessment was rated "unclear" (n = 41). The ratings were changed based on this practice for 31 studies (75.6% response rate).

Data synthesis

A series of random effects meta-analyses were performed using data from a subset of studies identified by the literature search performed by ourselves and Ohly and colleagues (2016). A common variance component was not assumed across studies or subgroups. For studies with more than one outcome measure, it was noted, where possible, which study was performed first at post-test, as these outcomes will be more sensitive to the restoration effect. For studies with multiple outcome measures from the same participant in the same meta-analysis, the variance was adjusted assuming non-independence and was represented as combined effects. Studies were selected based on the main inclusion criterion, namely, a measurement of baseline performance.

To begin, Ohly et al. (2016) method of comparing post-test outcome data between natural environments or stimuli (experimental) with non-natural environments or stimuli (control) was followed. Here, pooled mean differences were calculated by extracting means, SD, and sample sizes for each environmental condition and each outcome measure. Outcomes that measured conceptually similar cognitive constructs across different tasks were then grouped. This was done to obtain standardized mean effect size estimates (Hedges' g) for separate cognitive domains, as a platform for discussing the role of directed attention within each domain. Finally, to address the challenges noted by Ohly et al. (2016), moderator analyses were performed to determine the influence of three methodological variables on the pooled effect size. Within each cognitive domain, studies were grouped by baseline balance (yes, partial, unsure, no); restoration potential (heightened, normal); and exposure type (actual, virtual). Meta- and moderator analyses were conducted using Comprehensive Meta-Analysis (CMA 3.3.070).

Results

Part one - the systematic literature search

After searching 6 databases, 3997 results were identified that were screened for inclusion in our review. From this initial number, a total of 42 publications were included in the final review (Figure 2). Three publications reported results from multiple studies that were suitable for our analysis. In total, there were a total of 46 separate studies included in our final review.

It is worth noting the increasing productivity in the area of attention restoration research. Although our systematic literature search covered only 52 months between July 2013 and November 2017, there was a considerably larger number of relevant studies published during this time than in the years prior to July 2013. Figure 3 represents this increase in productivity graphically, where the red line marks the time at which Ohly et al. (2016) conducted their systematic search. All studies represented to the left of the line (n = 24) were included in Ohly and colleagues (2016) review, and all studies represented to the right of the line (n = 42) were included in the current review.

Study characteristics

Table presents the characteristics of included studies and is segmented by type of study. The majority (82.6%) of the 46 included studies utilized a randomized-controlled trial design. Of these, 29 studies used a between-subjects design where each subject was exposed only one environmental treatment and 9 studies used a within-subjects crossover design where each participant was exposed to both or all environmental treatments. There were 5 natural experiments that tested whether exposure to natural environments

Table 2. Quality appraisal of included studies.

	Berry et al. (2014)	Berry et al. (2015)	Beute and De Kort (2014), (1)	Beute and De Kort (2014), (2)	Bratman et al. (2015)	Chow and Lau (2015), (1)	Chow and Lau (2015), (2)	Chow and Lau (2015), (3)	Craig et al. (2015)	Dadvand et al. (2015)	Dadvand et al. (2017)	Emfield and Neider (2014)	Evensen et al. (2015)	Ferraro (2015)	Gamble, Howard, and Howard (2014)
Quality indicators															
Study design															
Power calculation reported	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No
Inclusion/exclusion criteria reported	No	Yes	No	No	Yes	No	No	No	No	Yes	No	Yes	No	No	No
Individual level allocation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	n/a	Yes	Yes	No	Yes
Random allocation to groups/ condition/order	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	n/a	Yes	Yes	No	Yes
Randomization procedure appropriate	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	n/a	Yes	Yes	No	Yes
Confounders															
Groups similar (sociodemographic)	**Yes**	Yes	**Yes**	**Yes**	Yes	Un.	Un.	Un.	**Yes**	n/a	n/a	Yes	Un.	**Yes**	Pa.
Groups balanced at baseline (cognitive scores)	Un.	Un.	Un.	Un.	Yes	Un.	Un.	Un.	Yes	n/a	n/a	Yes	Yes	Yes	**Pa.**
Participants blind to research question	Yes	Yes	**Yes**	**Yes**	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Intervention integrity															
Demonstrated need for attention restoration	No	No	Yes	Yes	No	Yes	Yes	Yes	No	n/a	n/a	No	Yes	No	No
Clear description of intervention and control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	n/a	Yes	Yes	No	Yes
Consistency of intervention (within and between groups)	Yes	Yes	Yes	Yes	Pa.[p]	No	Yes	Yes	Yes	n/a	n/a	Yes	Yes	No	Yes
Data collection methods															
Outcome assessors blind to group allocation	**No**	**No**	**No**	**No**	Un.	Un.	Un.	Un.	**No**	n/a	n/a	**Yes**	Un.	**Yes**	**No**
Baseline attention measures taken before the exposure	No	No	No	No	Yes	No	Yes	No	Yes	n/a	n/a	Yes	No	No	Yes
Consistency of data collection	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyses															
All attention outcomes reported (means and SD/SE)	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
All participants accounted for (i.e., losses/exlusions)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ITT analysis conducted (all data included after allocation)	No	No	No	No	Yes	Yes	Yes	Yes	Yes	n/a	n/a	No	Yes	Yes	Yes
Individual level analysis	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Statistical analysis methods appropriate for study design	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
External validity															
Sample representative of target population	**No[a]**	**No[a]**	**No[a]**	**No[a]**	Un.	No[a]	No[a]	Un.	No[a]	Yes	Yes	No[a]	No[a]	No[a]	**Pa.**
Overall quality score															
Total number of points	24/40	26/40	26/40	24/40	29/40	24/40	28/40	26/40	30/40	16/18	14/18	32/40	28/40	20/40	29/40
Quality rating as %	60	65	65	60	72.5	60	70	65	75	88.9	77.8	80	70	50	72.5
Responded to query about "uncertain" ratings	Yes	Yes	Yes	Yes	No	No	No	No	Yes	n/a	n/a	Yes	No	Yes	Yes
Criteria for meta-analysis															
Fatigue task	No	No	Yes	Yes	No	Yes	Yes	Yes	No	n/a	n/a	No	Yes	No	No
Baseline	No	No	No	No	Yes	No	Yes	No	Yes	n/a	n/a	Yes	Yes[b]	No	Yes

	Geniole et al. (2016)	Greenwood and Gatersleben (2016)	Haga et al. (2016)	Han (2017)	Jenkin et al. (2017)	Johansson et al. (2015)	Jung et al. (2017)	Kelz, Evans, and Röderer (2015)	Lee et al. (2015)	Li and Sullivan (2016)	Lin et al. (2014)	Lobo et al., (2015)	Lymeus, Lundgren, and Hartig (2017)	Rider and Bodner (2016) (1)	Rider and Bodner (2016)(2)
Quality indicators															
Study design															
Power calculation reported	No	No	No	No	No	No	Yes	No	No	No	No	No	Yes	No	No
Inclusion/exclusion criteria reported	Pa.	No	No	Yes	No	Yes	Yes	No	No	Yes	Pa.	No	Yes	No	No
Individual level allocation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Random allocation to groups/ condition/order	**Yes**	Yes	Yes	Yes	Yes	Pa.	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Randomization procedure appropriate	**Yes**	Yes	Yes	Yes	Yes	Yes	Yes	n/a	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Confounders															
Groups similar (sociodemographic)	Yes	**Yes**	Un.	Yes	Un.	Yes	Pa.	Yes[d]	**Yes**	Yes	**Yes**	**Un.**	**Yes**	**Pa.**	**Pa.**
Groups balanced at baseline (cognitive scores)	**Un.**	**Un.**	Un.	**Yes**	**Yes[o]**	**Yes**	Pa.	No	Yes	Yes	**Yes**	Un.	Yes	Un.	Un.
Participants blind to research question	Yes	Yes	Un.	**Yes**	**No**	**Yes**	**Yes**	Yes	**Yes**	Yes	Yes	Yes	**Yes**	Yes	Yes
Intervention integrity															
Demonstrated need for attention restoration	No	Yes	Yes	No	No[m]	Pa.[c]	Pa.[n]	n/a	No	Yes	No	Yes	Pa.[f]	No	No
Clear description of intervention and control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Consistency of intervention (within and between groups)	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Data collection methods															
Outcome assessors blind to group allocation	**No**	**Yes**	Un.	**No**	Yes	No	**Yes**	**No**	**No**	Un.	**No**	**Yes**	**No**	**No**	**No**
Baseline attention measures taken before the exposure	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No
Consistency of data collection	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyses															
All attention outcomes reported (means and SD/SE)	No	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes
All participants accounted for (i. e., losses/exlusions)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ITT analysis conducted (all data included after allocation)	Yes	Yes	Yes	No	No	No	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Individual level analysis	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Statistical analysis methods appropriate for study design	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
External validity															
Sample representative of target population	No[a]	**Yes**	No[a]	No[a]	No	**Yes**	**Yes**	**Yes**	No[a]	Un.	**No[a]**	No[a]	No[a]	**No[a]**	**No[a]**
Overall quality score															
Total number of points	27/40	34/40	26/40	28/40	26/40	30/40	37/40	24/36	30/40	32/40	31/40	26/40	31/40	25/40	25/40
Quality rating as %	67.5	85	65	70	65	75	92.5	66.7	75	80	77.5	65	77.5	62.5	62.5

(Continued)

Table 2. (Continued).

	Geniole et al. (2016)	Greenwood and Gatersleben (2016)	Haga et al. (2016)	Han (2017)	Jenkin et al. (2017)	Johansson et al. (2015)	Jung et al. (2017)	Kelz, Evans, and Röderer (2015)	Lee et al. (2015)	Li and Sullivan (2016)	Lin et al. (2014)	Lobo et al., (2015)	Lymeus, Lundgren, and Hartig (2017)	Rider and Bodner (2016) (1)	Rider and Bodner (2016)(2)
Responded to query about "uncertain" ratings	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Criteria for meta-analysis															
Fatigue task	No	Yes	Yes	No	No	Pa[c]	Pa.[n]	n/a	No	Yes	No	Yes	Pa.[f]	No	No
Baseline	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No

	Rogerson et al. (2016)	Sahlin et al., (2016)	)Schutte, Torquati, and Beattie (2017)	Sonntag-Öström et al., (2014)	Studente, Seppala, and Sadowska (2016)	Tanaka et al., (2013)	Triguero-Mas et al. (2017)	Ulset et al. (2017)	Van Den Berg et al. (2016)	Van Rompay and Jol (2016)	Wang et al., (2016)	Wang et al. (2017)	Ward et al. (2016)	Zhang, Kang, and Kang (2017)	Zijlema et al. (2017)
Quality indicators															
Study design															
Power calculation reported	No	No	No	Yes	No	No	Yes	No	No	No	No	No	No	No	No
Inclusion/exclusion criteria reported	No	No	Yes	Yes	No	Yes	Yes	No	Yes	No	Yes	No	Yes	No	No
Individual level allocation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	No	Yes	Yes	Yes	n/a	Yes	n/a
Random allocation to groups/condition/order	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	No	Yes	Yes	Yes	n/a	No	n/a
Randomization procedure appropriate	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	n/a	Yes	Yes	Yes	n/a	No[l]	n/a
Confounders															
Groups similar (sociodemographic)	Yes	Yes	Yes	Yes	**Yes**	Un.	Yes	n/a	Yes	**Yes**	**Yes**	**Yes**	n/a	Un.	n/a
Groups balanced at baseline (cognitive scores)	Yes	**Un.**	**Un.**	Un.	**Yes**	Yes	**Yes[o]**	n/a	Yes[o]	**No**	**Yes**	Yes	n/a	Un.	n/a
Participants blind to research question	Yes	Yes	Yes	Yes	Yes	Yes	**Yes**	Yes	Yes	Yes	Yes	Yes	Yes	Un.	Yes
Intervention integrity															
Demonstrated need for attention restoration	No	No	Yes	Yes[g]	No	Yes	No	n/a	n/a	No	Yes[j]	Yes[k]	n/a	Yes	n/a
Clear description of intervention and control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	n/a	Yes	Yes	Yes	No	n/a	Yes	n/a
Consistency of intervention (within and between groups)	Yes	No	Yes	Yes	Yes	Yes	Yes	n/a	Yes	Yes	Yes	Yes	n/a	Pa.	n/a
Data collection methods															
Outcome assessors blind to group allocation	**No**	**Un.**	**Pa.**	**No**	**Yes[h]**	Un.	No	n/a	Un.	**Yes[h]**	**Yes**	No	n/a	Un.	n/a
Baseline attention measures taken before the exposure	Yes	Yes	No	No	No	Yes	Yes	n/a	Yes	No	Yes	Yes	n/a	Yes	n/a
Consistency of data collection	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyses															
All attention outcomes reported (means and SD/SE)	Yes	Yes	**Yes**	Yes	No	No	No	Yes	Yes	Pa.	Yes	Pa.	No	Pa.	Yes

(*Continued*)

Table 2. (Continued).

	Rogerson et al. (2016)	Sahlin et al., (2016)	)Schutte, Torquati, and Beattie (2017)	Sonntag-Öström et al., (2014)	Studente, Seppala, and Sadowska (2016)	Tanaka et al., (2013)	Triguero-Mas et al. (2017)	Ulset et al. (2017)	Van Den Berg et al. (2016)	Van Rompay and Jol (2016)	Wang et al., (2016)	Wang et al. (2017)	Ward et al. (2016)	Zhang, Kang, and Kang (2017)	Zijlema et al. (2017)
All participants accounted for (i.e., losses/exlusions)	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ITT analysis conducted (all data included after allocation)	Yes	Yes	Yes	No	No	Yes	Yes	n/a	No	Yes	Yes	Yes	n/a	No	n/a
Individual level analysis	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Statistical analysis methods appropriate for study design	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
External validity															
Sample representative of target population	**Un.**	**Un.**	**Yes**	**Yes**	No[a]	Un.	Yes	Yes	Un.	**Yes**	No[a]	No[a]	**Un.**	No[a]	Yes
Overall quality score															
Total number of points	30/40	26/40	33/40	32/40	24/40	30/40	34/40	14/18	24/36	29/40	36/40	29/40	12/18	18/40	14/18
Quality rating as %	75	65	82.5	80	60	75	85	77.8	66.7	72.5	90	72.5	66.7	45	77.8
Responded to query about "uncertain" ratings	Yes	Yes	Yes	Yes	Yes	No	Yes	n/a	No	Yes	Yes	Yes	Yes	No	n/a
Criteria for meta-analysis															
Fatigue task	No	No	Yes	Yes[g]	No	Yes	No	n/a	n/a	No	Yes[j]	Yes[k]	n/a	Yes	n/a
Baseline	Yes	Yes	No	No	No	Yes	Yes	n/a	Yes	No	Yes	Yes	n/a	Yes	n/a

Notes. Total number of possible points: 40 for RCT design; 36 for quasi-experimental design; 18 for natural experiment design.
[a]No target population per se, however, participants were a convenience sample of university students.
[b]Baseline measures, fatigue task, and post measures all completed during exposure of natural/non-natural stimuli.
[c]Not experimentally induced. Participants reported mental fatigue.
[d]Education level of parents differed between groups. This was controlled for in the statistical analysis, with no influence on effect.
[e]Single outcome measure derived from two tasks (DSF + DSB).
[f]Not experimentally induced. Study took place at the end of a workday.
[g] Not experimentally induced. Participants recruited based on diagnosis of Exhaustion Disorder.
[h]Particularly important for the subjective nature of outcome measures.
[i]ATMT contains subtasks with varying difficulty level. Combined scores analyzed and reported.
[j]Participants gave a speech in English. Used to evoke stress response rather than fatigue attention.
[k]Depletion used as between-subjects factor.
[l]Participants not randomized, but matched on gender and grade of a recent test.
[m]An attempt was made to induce 'ego depletion' but that task was not suitable for the participants, likely due to age.
[n]Participants were patients who had suffered heart failure.
[o] Baseline used as covariate in the linear regression model.

(commonly referred to as greenspace in these studies) influenced cognitive development over longer time periods. Three quasi-experiments were included where assignment to environmental exposure could not be randomized. In two cases (Kelz, Evans, and Röderer 2015; Van Den Berg et al. 2016) these studies took place in school settings, while the remaining study examined the effects of a wilderness hike (Ferraro 2015).

The included studies were conducted largely in western countries. Twenty studies took place in Europe (43.5%); 15 in North America (32.6%); one in Australia (2.2%); one in New Zealand (2.2%); and 9 in Asia (19.5%). One of the natural experiments involved data from multiple countries (Zijlema et al. 2017).

A range of age groups, from young children (m = 4.53 years; Schutte, Torquati, and Beattie 2017) to elderly (m = 69.1 years; Gamble, Howard, and Howard 2014), were represented in the sample populations across the included studies. However, young adults, represented mostly by university students, made up the largest proportion (52.2%). Most investigators recruited participants with typical cognitive function; however, several studies included participants where restoration potential was expected to be heightened. In three cases this was based on a specific diagnosis, including heart failure (Jung et al. 2017), acquired brain injury (Johansson et al. 2015), and exhaustion disorder (Sonntag-Ostrom et al. 2014). Other investigations included participants with presumed higher restoration potential due to prior lectures (Lin et al. 2014) or end-of-day testing sessions (Lymeus, Lundgren, and Hartig 2017). The final study to consider restoration potential based on participant characteristics was Craig and colleagues (2015), who grouped university students based on typical or heightened depressive symptom ratings.

The included studies used actual physical exposure (54.3%) or virtual exposure (45.7%) as the environmental treatments. Walking was the most frequently recorded type of engagement in studies that involved actual physical exposure. In some cases, walking was used in combination with another activity, such as taking photos (Bratman et al. 2015) and relaxing (Sonntag-Ostrom et al. 2014; Zhang, Kang, and Kang 2017); while one study instructed the participants to act in the environments as they normally would (Triguero-Mas et al. 2017). More active engagement was employed in some studies, including hiking (Ferraro 2015) and cycling (Rogerson et al. 2016). Passive viewing of actual natural environments or stimuli was included in randomized-controlled trials (Evensen et al. 2015; Greenwood and Gatersleben 2016; Li and Sullivan 2016; Studente, Seppala, and Sadowska 2016) and quasi-experimental studies (Van Den Berg et al. 2016). While natural experiments explored exposure to green space in residential areas (Dadvand et al. 2015, 2017) and/or school environments (Dadvand et al. 2015; Ward et al. 2016) Virtual exposure to environments was used in 55.3% of all randomized-controlled trials. Virtual exposure consistent almost exclusively of image or video viewing; however, sound was also investigated by Emfield and Neider (2014) as a between-subjects factor. Haga and colleagues (2016) explored expectation bias and stimuli-source attribution by telling participants who listened to white noise that they were hearing either a natural environment with a waterfall or an industrial environment with machinery. In some studies, virtual environments were supplemented with other engagement activities such as imagining presence in the environment (Chow and Lau 2015; Van Rompay and Jol 2016) or mindfulness meditation (Lymeus, Lundgren, and Hartig 2017).

The duration of exposure to environmental treatments ranged considerably between the studies. Randomized-controlled trials ranged from 'micro-breaks' of 40 s (Lee et al. 2015) to three-hr exposures (Triguero-Mas et al. 2017), while some included a course of exposures lasting several weeks (Johansson et al. 2015; Lymeus, Lundgren, and Hartig 2017). In the quasi-experimental studies and natural experiments, where exposure to natural environments or stimuli was of interest, durations were considerably longer, ranging from 6 days (Ferraro 2015) to several years (Dadvand et al. 2017; Ulset et al. 2017).

As reported in Table 2, the quality ratings of the included studies all fell within the moderate (34–66%) and high (67–100%) classifications used by Ohly et al. (2016). As Ohly et al. (2016) noted, the quality ratings judged publications on overall experimental quality as well as the extent to

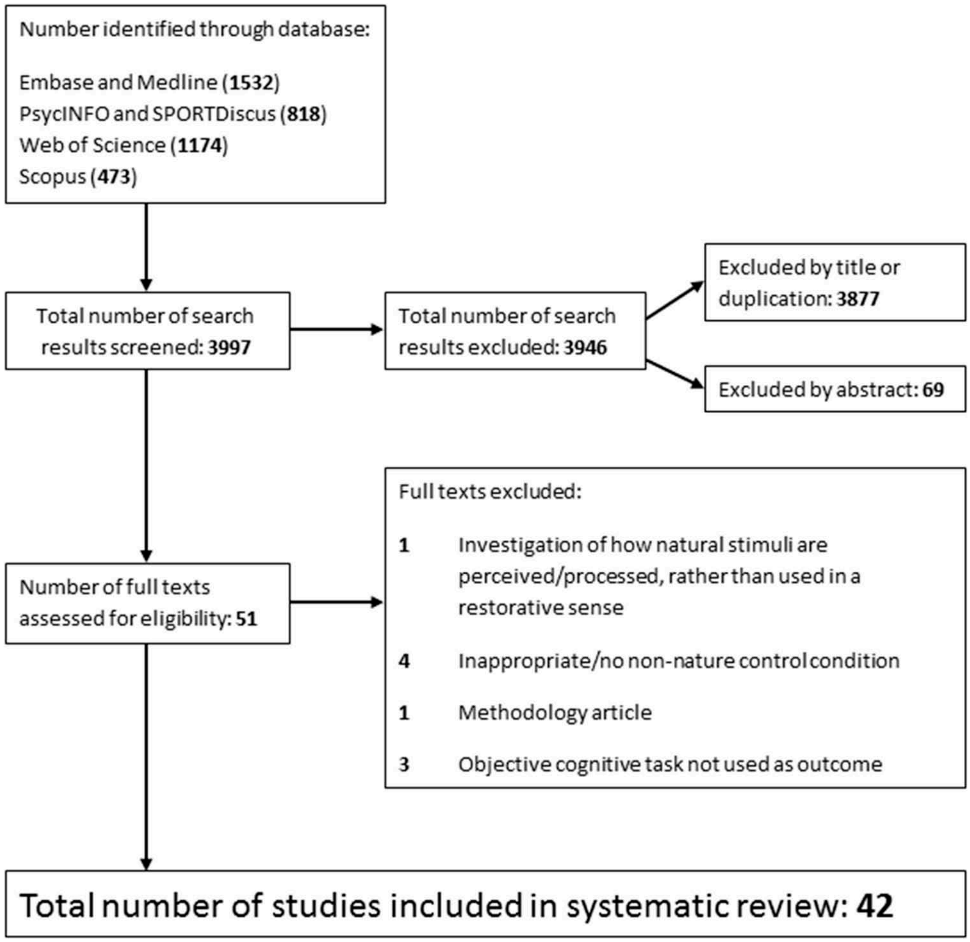

Figure 2. PRISMA flowchart of the screening procedure.

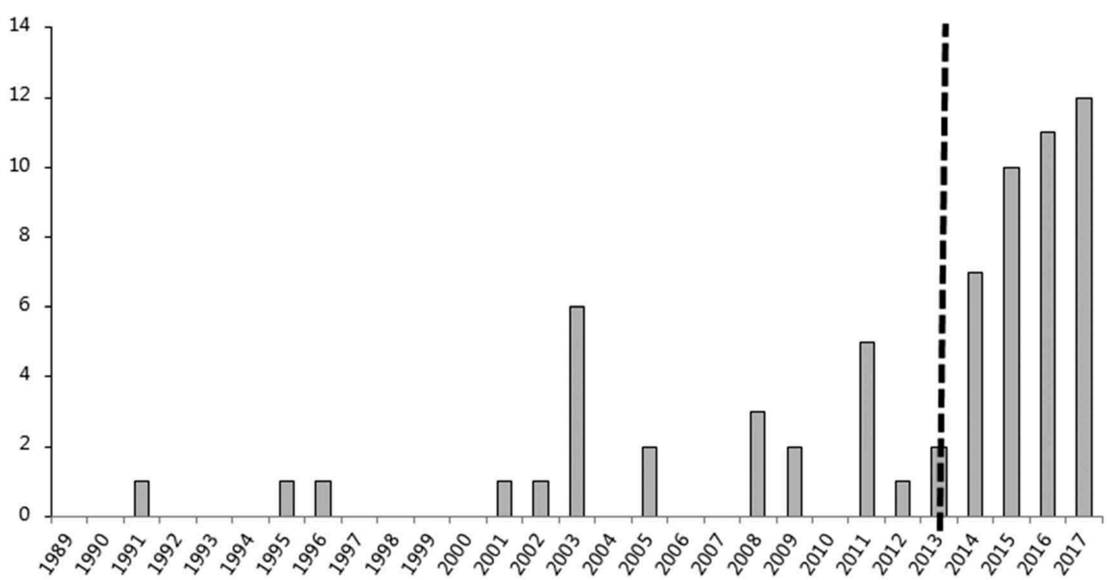

Figure 3. Timeline of attention restoration studies by year. The dashed line represents the time of Ohly et al.'s (2016) systematic literature search.

which related to the reviews on attention restoration research, which was not always the aim of the included experiments. Trends revealed by the quality assessment include a lack of power calculations, variety in the extent to which inclusion and exclusion criteria were reported, whether participants were blind to research hypotheses, and whether outcome assessors were blinded to group allocation during the assessment. An analysis of whether participants were representative of the target population was rarely analyzed or reported. In many cases, the target population was not explicit, though this is likely due to the frequency of university students recruited as participants.

Part two - meta-analysis: the effect of natural environments on cognitive performance

To synthesize the most relevant data, only studies that reported baseline measures of cognitive performance were included. Our results were supplemented with data from the studies identified by Ohly and colleagues (2016) that also included baseline measures of cognitive performance in order to conduct our meta-analyses. Across both reviews, data from a total of 19 different cognitive tasks were included, spanning 7 domains of cognition. A total of 49 separate outcome measures were derived from the cognitive tasks. One conglomerate score of attention performance was not included (Cimprich and Ronis 2003) as the results from its subtasks were included separately.

Results are presented in separate sections based on each of the 8 general cognitive domains assessed by the outcome measures identified in the review (i.e., working memory, attentional control, visual attention processes, vigilance, cognitive flexibility, impulse control, processing speed, and other emerging domains). First, general characteristics of the studies are presented, including means, SD and sample sizes, and characteristics of the moderators to be considered for analysis. The meta-analysis based on post-test scores is then reported as well as additional moderator analyses, where possible. For cognitive domain sections where meta-analysis was not appropriate, a narrative analysis is provided.

Working memory

A total of 7 outcome measures derived from 4 tasks were categorized as being relevant to the working memory domain of cognition (Table 3). These measures were grouped together as performance relied on the ability to hold and manipulate to-be-remembered information in the short-term (Baddeley 2003). The outcome measures can be differentiated by aspects such as their demand on other cognitive processes (e.g., Digit Span Forward vs. Digit Span Backward; Jung et al. 2017) or whether performance is based on verbal (e.g., Digit Span Backwards) or visuospatial memory processes (Forward Spatial Span; Han 2017).

As illustrated in Figure 4 and Table 4, natural environments or stimuli were found to have a significantly positive effect on working memory performance when effect sizes were pooled across all levels of moderators (n = 22), with a pooled estimated mean effect size (Hedges' g) of 0.162. As seen in Figure 4B (supplementary material), the positive effect remains (Hedges' g = 0.172) when only including studies with reported baseline balance (n = 10). A significant difference was found between studies in which participants' restoration potential was heightened and studies where restoration potential was normal. As Figure 4C (supplementary material) demonstrates, studies that included participants with heightened restoration potential reported a higher pooled effect size (Hedges' g = 0.307) than those with normal restoration potential (Hedges' g = 0.052). Studies that used actual exposure to real environments showed significant positive effects (Hedges' g = 0.156) in favor of natural environments, (n = 13) as presented in Figure 4D (supplementary material). However, there was no significant difference in effect sizes between actual and virtual exposures.

Attentional control

A total of 10 outcome measures derived from 4 cognitive tasks were categorized as being relevant to attentional control (see Table 5). This cognitive domain is sometimes referred to as executive attention, focused attention, selective attention, interference control, and inhibition at the level of attention (Diamond 2013). A key component of these tasks is the need to direct attention towards a target while simultaneously ignoring distractors. Therefore, these measures may be most directly related to the effortful type of attention described

Table 3. Data and moderator details from included studies using outcome measures classified in the working memory domain.

			Natural environment/stimuli			Control environment/ stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
DSB (span score)	Berman, Jonides, and Kaplan (2008) (1)	-	37	7.90 (2.28)	9.40 (2.49)	37	7.90 (1.85)	8.4 (2.01)	task	35⊠00″	T1	yes	7.90	actual
	Berman, Jonides, and Kaplan (2008) (2)	-	12	7.92 (3.33)	9.33 (2.98)	12	7.83 (3.60)	8.83 (3.12)	none	-	T2†	partial	7.88	virtual
	Berman et al. (2012)	-	19	7.42 (3.00	8.63 (2.87)	19	8.26 (2.51)	7.84 (2.24)	participant + task	NR	T1	no	7.84	actual
	Cimprich and Ronis (2003)	-	83	4.99 (1.37)	5.20 (1.28)	74	4.51 (1.12)	4.58 (1.20)	participant	-	T2*	no	4.75	actual
	Emfield and Neider (2014)	sound	28	5.3 (1.44)	6.11 (1.42)	28	5.75 (1.04)	5.79 (1.23)	none	-	T2†	yes	5.53	virtual
		pictures	27	5.28 (0.98)	5.52 (1.33)	27	5.64 (1.22)	5.60 (1.23	none	-	T2†	yes	5.46	virtual
		sound + pictures	28	5.40 (1.16)	5.88 (1.20)	27	5.52 (1.16)	5.44 (1.25)	none	-	T2†	yes	5.46	virtual
	Gamble, Howard, and Howard (2014)	older adults	15	5.78 (1.72)	7.11 (1.69)	15	6.09 (2.02)	7.00 (2.61)	none	-	T2†	partial	5.94	virtual
		younger adults	13	7.91 (2.43)	8.27 (2.61)	13	7.56 (2.60)	9.33 (2.65)	none	-	T2†	partial	7.74	virtual
	Han (2017)	collapsed over activity conditions	58	9.84 (2.85)	10.52 (2.62)	58	9.83 (3.11)	10.67 (2.50)	none	-	T2†	yes	9.84	actual
	Jung et al. (2017)	healthy adults	20	5.05 (0.95)	5.40 (1.57)	20	4.55 (0.89)	5.00 (1.45)	none	-	T2†	partial	4.80	virtual
		heart failure patients	20	4.40 (1.54)	4.55 (1.79)	20	4.30 (1.75)	4.85 (1.42)	participant	-	T2*	partial	4.35	virtual
	Lin et al. (2014)	collapsed over awareness conditions	104	6,43 (1.53)	7,31 (1,47)	34	7.12 (1.45)	6.53 (1.38)	participant	-	T2*	yes	6.78	virtual
	Stark (2003)	-	29	5.1 (1.3)	5.4 (1.6)	28	4.7 (1.1)	5.0 (1.6)	participant	-	T2*	partial	4.90	actual
DSB (total correct/ 14)														
	Triguero-Mas et al. (2017)	after 30″ exposure	52	4,54 (2,5)	5,25 (2,5)	26	5.21 (2.80)	5.65 (2.66)	none	-	T2†	yes	4.88	actual
		after 180″ exposure	52	4,54 (2,5)	5,37 (2,68)	26	5.21 (2.80)	5.88 (2.83)	none	-	T2†	yes	4.88	actual
DSB (total correct/9)	Rogerson et al. (2016)	-	24	3.2 (1.7)	3.5 (1.7)	24	3.5 (1.4)	3.0 (1.6)	none		T2†	yes	3.35	actual
DSF (span score)	Cimprich and Ronis (2003)	-	83	6.71 (1.26)	6.98 (1.37)	74	6.54 (1.12)	6.53 (1.38)	participant	-	T2*	no	6.63	actual
	Jung et al. (2017)	healthy adults	20	6.85 (1.18)	6.90 (1.41)	20	6.80 (1.15)	7.25 (1.21)	none	-	T2†	partial	6.83	virtual
		heart failure patients	20	6.80 (1.44)	6.55 (1.19)	20	6.70 (1.22)	6.55 (1.15)	participant	-	T2*	partial	6.75	virtual
	Stark (2003)	-	29	6.8 (1.2)	7.1 (1.4)	28	6.6 (1.0)	6.7 (1.1)	participant	-	T2*	partial	6.70	actual
FSS (span score)	Han (2017)	collapsed over activity conditions	58	9.26 (1.77)	9.97 (1.94)	58	8.78 (1.83)	9.47 (1.40)	none	-	T2†	yes	9.02	actual
RST	Evensen et al. (2015)	window view, T1 baseline	31	0.65 (0.11)	0.71 (0.10)	19	0.65 (0.11)	0.71 (0.10)	task	15⊠00″	T1	yes	0.65	actual
		window view, T2 baseline	31	0.69 (0.09)	0.71 (0.10)	19	0.69 (0.09)	0.71 (0.10)	task	15⊠00″	T2*	yes	0.69	actual

(*Continued*)

Table 3. (Continued).

Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	Natural environment/stimuli			Control environment/ stimuli			Moderators					Exposure
				Pre	Post		Pre	Post	Restoration potential		Baseline			
			n	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
		no window view, T1 baseline	31	0.64 (0.10)	0.69 (0.07)	19	0.64 (0.07)	0.72 (0.09)	task	15⊠00″	T1	yes	0.64	actual
		no window view, T2 baseline	31	0.66 (0.09)	0.69 (0.07)	19	0.68 (0.09)	0.72 (0.09)	task	15⊠00″	T2*	yes	0.67	actual
DSF + DSB (composite)	Bodin and Hartig (2003)	men	6	10.83 (2.32)	11.92 (2.31)	6	11.92 (2.76)	12.17 (2.89)	none	-	T2†	partial	11.38	actual
		women	6	11.67 (3.82)	10.58 (3.68)	6	10.67 (2.89)	11.25 (3.22)	none	-	T2†	partial	11.17	actual

Notes. DSB = Digit Span Backwards; DSF = Digit Span Forwards; FSS = Forward Spatial Span; RST = Reading Span Task; *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

in ART. Three of these tasks (Stroop, Attention Network Task (ANT), and Multi-Source Interference Task (MSIT) involve two trial types, congruent and incongruent. Incongruent trials involve targets and distractors designed to evoke cognitive conflict that must be overcome using attentional control processes. Recruitment of these processes is reflected in response times to such trials being typically slower. Conversely, congruent trials involve harmonious targets and distractors where responses reflect cognitive processes without this conflict resolution. Optimally, outcome measures take both trial types into account when calculating attentional control scores to adjust for general processing speed.

As illustrated by Figure 5 and Table 6, natural environments or stimuli were found to exert a significant positive effect on attentional control when effect sizes were pooled across all levels of moderators (n = 17), with a pooled estimated mean effect size (Hedges' g) of −0.156. As Figure 5B (supplementary material) shows, the significant effect was lost when only including studies with reported baseline balance (n = 7). Figure 5C (supplementary material) illustrates that no significant differences were found in relation to restoration potential. As noted in Figure 5D, investigations that used actual exposure to real environments exhibited significantly positive effects (Hedges' g = −2.612) in favor of natural environments (n = 7); however, there was no significant difference in effect sizes between actual and virtual exposures.

Visual attention

Seven outcome measures derived from two cognitive tasks were categorized as being related to visual processing (Table 7). Six of the outcome measures were derived from the Functional Field of View Task (Emfield and Neider 2014), which measures how distance and placement of a target stimulus from a fixation point affect speed and accuracy of short-term spatial recall. Thus, the task is believed to indicate the spatial area in which stimuli can be attended to (Ball et al. 1988). The final outcome measure was the orienting score from the ANT, which assesses the ability

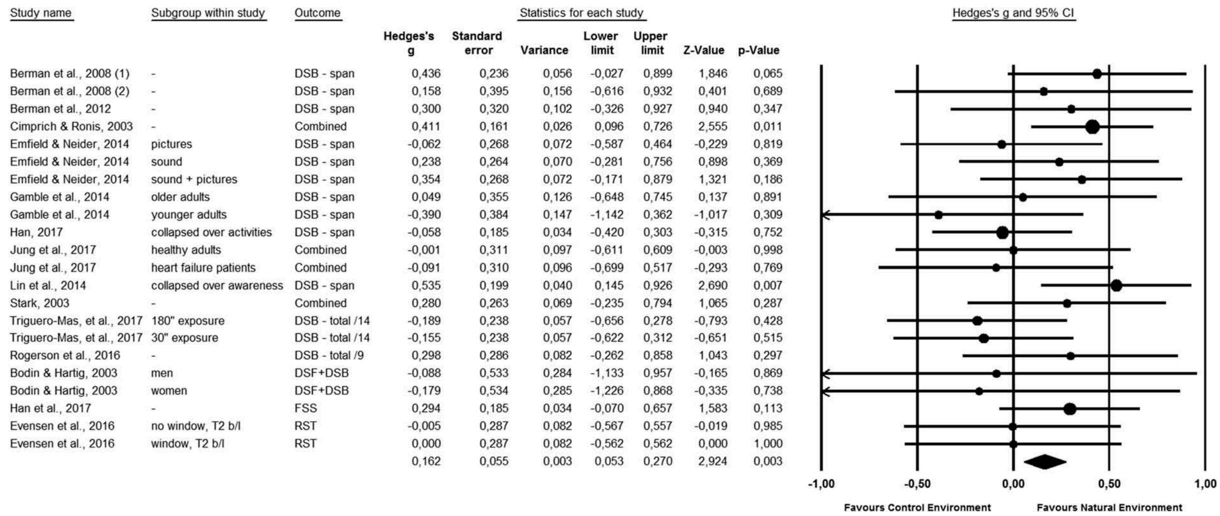

Figure 4. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to working memory. No moderator variables are included.

Table 4. Summary of results from the working memory meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Working Memory					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	22	0.162 (0.053 – 0.270)	0.00	2.924	**0.003**	-
Moderators						
Baseline balance						Q = 4.535, DF = 2, p = 0.104
yes	10	0.172 (0.005 – 0.339)	26.37	2.020	**0.043**	
partial	10	0.010 (−0.200 – 0.219)	0.00	0.090	0.928	
no	2	0.389 (0.107 – 0.670)	0.00	2.705	**0.007**	
unsure	-	-	-	-	-	
Restoration potential						Q = 5.211, DF = 1, p = **0.022**
heightened	8	0.307 (0.142 – 0.472)	0.00	3.641	**0.000**	
normal	14	0.052 (−0.092 – 0.195)	0.00	0.710	0.478	
Exposure type						Q = 0.018, DF = 1, p = 0.894
actual	13	0.156 (0.024 – 0.289)	0.00	2.310	**0.021**	
virtual	9	0.172 (−0.016 – 0.360)	0.00	1.797	0.072	

to select visual information from sensory input (Fan et al. 2002).

As illustrated in Figure 6 and Table 8, natural environments or stimuli were found to exert no marked effect on visual attention measures when effect sizes were pooled across all levels of moderators. Baseline balance was the only moderator that had sufficient data across at least two levels. However, as Figure 6B (supplementary material) demonstrates, no marked effect of environment for studies with full ($n = 3$) or partial ($n = 3$) baseline balance was found.

Vigilance

A total of 7 outcome measures derived from three cognitive tasks were categorized as being relevant to vigilance (Table 9). This cognitive domain is sometimes referred to as sustained attention and describes tasks where participants need to maintain alertness over longer periods of time in order to detect irregular or unpredictable events (Sarter, Givens, and Bruno 2001). Outcome measures derived from performance requiring a simple response over an extended time were included in this cognitive domain. Most outcome measures were taken from the Sustained Attention to Response Task, where participants need to respond to all stimuli except a target; however congruent trials of the MSIT (Jung et al. 2017) and the alerting score of the ANT (Berman, Jonides, and Kaplan 2008; Gamble, Howard, and Howard 2014) were also considered relevant. Data

Table 5. Data and moderator details from included studies using outcome measures classified in the attentional control domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
NCPC (number of reversals[a])	Greenwood and Gatersleben (2016)	collapsed over context conditions	60	5.76 (1.99)	3.84 (1.69)	60	5.32 (2.32)	4.59 (2.46)	task	17″00	T2	unsure	5.54	Actual
	Hartig et al. (2003)	post test during exposure, no fatigue	28	4.17 (1.92)	4.06 (1.90)	28	3.70 (1.35)	4.43 (1.67)	none	-	T2†	yes	3.94	Actual
		post test after exposure, no fatigue	28	4.17 (1.92)	3.88 (2.21)	28	3.70 (1.35)	4.09 (1.72)	none	-	T2†	yes	3.94	Actual
		post test during exposure, fatigue	28	4.37 (1.92)	3.96 (1.93)	28	3.87 (1.85)	4.76 (2.19)	task	40″00	T1	yes	4.12	Actual
		post test after exposure, fatigue	28	4.37 (1.92)	3.98 (2.44)	28	3.87 (1.85)	4.67 (2.63)	task	40″00	T1	yes	4.12	Actual
NCPC (number of reversals[b])	Sahlin et al. (2016)	-	51	5.27 (2.25)	4.78 (2.74)	51	5.73 (2.82)	5.61 (3.63)	none	-	T2†	unsure	5.50	Actual
NCPC (% reduction[c])	Cimprich and Ronis (2003)	-	83	10.29 (60.22)	19.48 (37.81)	74	17.87 (66.50)	13.87 (62.88)	participants	-	T2*	no	14.08	Actual
ANT (executive score)	Berman, Jonides, and Kaplan (2008) (2)	-	12	86 (39.14)	67 (29.27)	12	81 (53.69)	93 (62.22)	none	-	T2†	partial	83.50	Virtual
	Kelz, Evans, and Röderer (2015)	-	72	137.25 (42.25)	104.79 (34.61)	61	165.14 (65.74)	114.00 (33.58)	none	-	T2†	no	151.20	Actual
	Gamble, Howard, and Howard (2014)	older adults	15	146.77 (59.53)	101.65 (51.74)	15	103.07 (47.45)	95.22 (41.87)	none	-	T2†	partial	124.92	Virtual
		younger adults	13	130.40 (37.33)	104.05 (37.93)	13	106.07 (54.88)	104.03 (54.61)	none	-	T2†	partial	118.24	Virtual
	Emfield and Neider (2014)	sound	28	105 (36.0)	102 (35.7)	28	118 (36.8)	107 (35.5)	none	-	T2†	yes	111.50	Virtual
		pictures	27	110 (52.6)	101 (52.2)	27	113 (32.9)	106 (34.7)	none	-	T2†	yes	111.50	Virtual
		sound + pictures	28	117 (54.7)	124 (49.9)	27	114 (41.5)	110 (30.4)	none	-	T2†	yes	115.50	Virtual
Stroop Task (% error, incongruent)	Jung et al. (2017)	healthy adults	20	11.70 (9.82)	4.24 (4.38)	20	10.82 (10.47)	5.70 (7.26)	none	-	T2†	partial	11.26	Virtual
		heart failure patients	20	23.61 (18.28)	18.19 (18.30)	20	19.72 (15.81)	14.31 (16.03)	participants	-	T2*	partial	21.67	Virtual
Stroop Task (RT, incongruent)	Jung et al. (2017)	healthy adults	20	1630 (510)	1445 (467)	20	1595 (522)	1407 (423)	none	-	T2†	partial	1612.50	Virtual
		heart failure patients	20	1641 (541)	1616 (513)	20	1598 (353)	1659 (387)	participants	-	T2*	partial	1619.50	Virtual

(*Continued*)

Table 5. (Continued).

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
Stroop Task (total correct, incongruent)	Jenkin et al. (2017) (1)	-	25	45.68 (3.21)	47.44 (2.52)	25	44.76 (3.64)	46.20 (3.38)	none	-	T2†	yes	45.22	Virtual
	Jenkin et al. (2017) (2)	-	26	40.12 (11.48)	42.65 (9.58)	28	37.43 (11.34)	39.82 (10.92)	none	-	T2†	yes	38.78	Virtual
Stroop Task (interference score)	Geniole et al. (2016)	-	31	−51.31 (80.67)	−24.45 (71.42)	31	−44.50 (70.43)	−29.95 (57.88)	none	-	T2†	unsure	−47.91	Actual
MSIT (% error, incongruent)	Jung et al. (2017)	healthy adults	20	4.13 (4.00)	1.77 (1.99)	20	3.67 (3.93)	2.13 (2.20)	none	-	T2†	partial	3.90	Virtual
		heart failure patients	20	4.79 (3.40)	4.59 (4.42)	20	4.67 (2.42)	4.27 (3.92)	participants	-	T2*	partial	4.73	Virtual
MSIT (RT, incongruent)	Jung et al. (2017)	healthy adults	20	921 (148)	890 (129)	20	921 (152)	886 (154)	none	-	T2†	partial	921	Virtual
		heart failure patients	20	1047 (150)	1019 (145)	20	1043 (123)	1015 (128)	participants	-	T2*	partial	1045	Virtual

Notes. NCPC = Necker Cube Pattern Control; ANT = Attention Network Task; MSIT = Multi-Source Interference Task. *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

[a]Average of 2x30s periods.

[b]One 30 s period.

[c]From baseline to holding conditions.

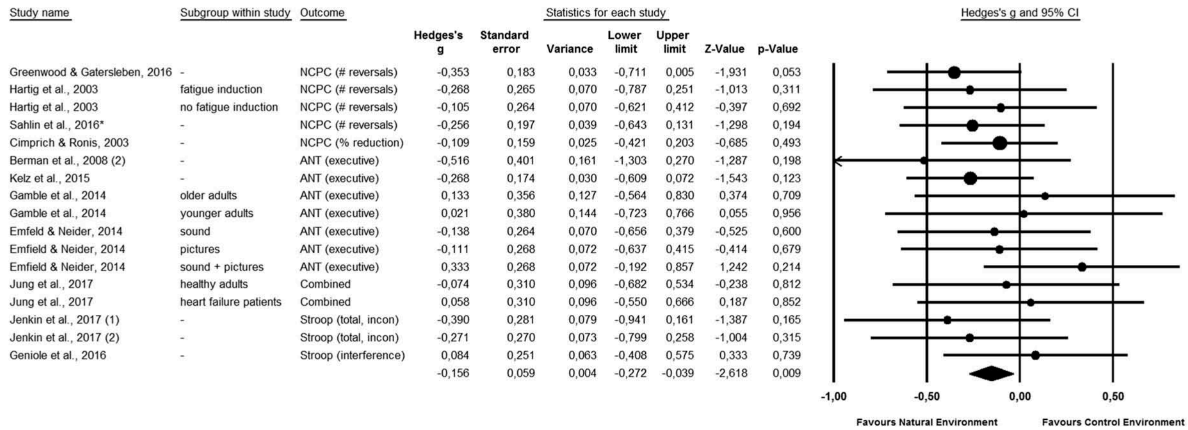

Figure 5. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to attentional control. No moderator variables are included.

Table 6. Summary of results from the attentional control meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Attentional Control					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	17	−0.156 (−0.272 – 0.039)	0	−2.618	**0.009**	-
Moderators						
Baseline balance						Q = 0.848, DF = 3, p = 0.838
yes	7	−0.132 (−0.331 – 0.066)	0	−1.306	0.191	
partial	5	−0.052 (−0.355 – 0.251)	0	−0.337	0.736	
no	2	−0.182 (−0.412 – 0.048)	0	−1.548	0.122	
unsure	3	−0.221 (−0.454 – 0.013)	1.33	−1.852	0.064	
Restoration potential						Q = 0.191, DF = 1, p = 0.662
heightened	4	−0.192 (−0.395 – 0.010)	0	−1.866	0.062	
normal	13	−0.137 (−0.280 – 0.005)	0	−1.888	0.059	
Exposure type						
actual	7	−0.199 (−0.348 – −0.050)	0	−2.612	**0.009**	Q = 0.825, DF = 1, p = 0.364
virtual	10	−0.088 (−0.275 – 0.098)	0	−0.926	0.355	

for a well-known outcome measure in vigilance research, d-prime, which measures the ability to discriminate targets from other stimuli, was only available from one publication (Berto 2005).

As shown in Figure 7 and Table 10, natural environments or stimuli were found to exert no marked effect on vigilance when effect sizes were pooled across all levels of moderators (n = 8). Due to a lack of sufficient data across more than two levels of baseline balance and restoration potential, moderator analyses were not performed. The influence of exposure type could not be explored as all studies used virtual exposures.

Impulse control

Two outcome measures derived from one cognitive task were categorized as being relevant to impulse control (Table 11). This cognitive domain is related to self-control and response inhibition, sometimes termed behavioral inhibition, and refers to the ability to control behavior rather than thought or attentional processes (Diamond 2013). Outcome measures were categorized as assessing impulse control if they assess the ability to exert inhibition of a pre-potent response. The measures identified include errors of commission on the SART where the participant failed to override the pre-potent response tendency to respond to non-target stimuli and mistakenly responded to a target, for which the correct response is to withhold. The other outcome measures identified is the reverse score on the same task, where correct responses to targets were tallied so that a higher count reflects better impulse control.

Table 7. Data and moderator details from included studies using outcome measures classified in the visual attention domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					Exposure
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
ANT (orienting score)	Berman, Jonides, and Kaplan (2008) (2)	-	12	47 (22.38)	55 (25.39)	12	46 (34.68)	43 (16.39)	none	-	T2†	partial	46.50	virtual
	Gamble, Howard, and Howard (2014)	older adults	15	41.59 (32.43)	71.72 (38.70)	15	67.80 (53.88)	91.54 (123.77)	none	-	T2†	partial	54.70	virtual
		younger adults	13	64.12 (29.72)	50.67 (33.55)	13	35.78 (29.30)	45.18 (35.81)	none	-	T2†	partial	49.95	virtual
FFOV (RT, 10^0)	Emfield and Neider (2014)	sound	28	935 (253)	811 (232)	28	951 (240)	779 (204)	none	-	T2†	yes	943.00	virtual
		pictures	27	933 (250)	739 (200)	27	823 (246)	817 (288)	none	-	T2†	yes	878.00	virtual
		sound + pictures	28	961 (304)	817 (288)	27	909 (278)	740 (200)	none	-	T2†	yes	935.00	virtual
FFOV (accuracy, 10^0)	Emfield and Neider (2014)	sound	28	0.768 (0.309)	0.848 (0.151)	28	0.791 (0.204)	0.912 (0.113)	none	-	T2†	yes	0.780	virtual
		pictures	27	0.766 (0.232)	0.864 (0.151)	27	0.729 (0.316)	0.814 (0.249)	none	-	T2†	yes	0.748	virtual
		sound + pictures	28	0.746 (0.320)	0.846 (0.266)	27	0.828 (0.231)	0.896 (0.126)	none	-	T2†	yes	0.787	virtual
FFOV (RT, 20^0)	Emfield and Neider (2014)	sound	28	1018 (294)	834 (236)	28	941 (224)	790 (208)	none	-	T2†	yes	979.50	virtual
		pictures	27	959 (269)	743 (210)	27	949 (292)	804 (229)	none	-	T2†	yes	954.00	virtual
		sound + pictures	28	979 (300)	816 (279)	27	914 (263)	756 (219)	none	-	T2†	yes	946.50	virtual
FFOV (accuracy, 20^0)	Emfield and Neider (2014)	sound	28	0.651 (0.283)	0.759 (0.277)	28	0.703 (0.243)	0.849 (0.144)	none	-	T2†	yes	0.677	virtual
		pictures	27	0.681 (0.293)	0.824 (0.171)	27	0.659 (0.321)	0.754 (0.284)	none	-	T2†	yes	0.670	virtual
		sound + pictures	28	0.698 (0.311)	0.800 (0.275)	27	0.800 (0.225)	0.861 (0.148)	none	-	T2†	yes	0.749	virtual
FFOV (RT, 30^0)	Emfield and Neider (2014)	sound	28	1064 (381)	851 (258)	28	989 (249)	804 (210)	none	-	T2†	yes	1026.50	virtual
		pictures	27	989 (263)	795 (237)	27	999 (297)	834 (252)	none	-	T2†	yes	994.00	virtual
		sound + pictures	28	983 (317)	820 (308)	27	934 (274)	767 (249)	none	-	T2†	yes	958.50	virtual
FFOV (accuracy, 30^0)	Emfield and Neider (2014)	sound	28	0.543 (0.238)	0.662 (0.200)	28	0.535 (0.217)	0.627 (0.174)	none	-	T2†	yes	0.539	virtual
		pictures	27	0.511 (0.268)	0.619 (0.247)	27	0.524 (0.268)	0.608 (0.256)	none	-	T2†	yes	0.518	virtual
		sound + pictures	28	0.564 (0.306)	0.634 (0.284)	27	0.659 (0.149)	0.710 (0.147)	none	-	T2†	yes	0.612	virtual

Notes. ANT = Attention Network Task; FFOV = Functional Field of View. *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

Study name	Subgroup within study	Outcome	Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Berman et al., 2008 (2)	-	ANT (orienting)	0,542	0,402	0,161	-0,245	1,330	1,349	0,177
Gamble et al., 2014	older adults	ANT (orienting)	-0,210	0,356	0,127	-0,909	0,488	-0,591	0,555
Gamble et al., 2014	younger adults	ANT (orienting)	0,153	0,380	0,145	-0,592	0,899	0,403	0,687
Emfield & Neider, 2014	pictures	Combined	-0,220	0,269	0,072	-0,747	0,308	-0,816	0,415
Emfield & Neider, 2014	sound	Combined	0,205	0,265	0,070	-0,314	0,723	0,773	0,439
Emfield & Neider, 2014	sound + pictures	Combined	0,260	0,267	0,071	-0,263	0,784	0,975	0,330
			0,100	0,126	0,016	-0,147	0,347	0,793	0,427

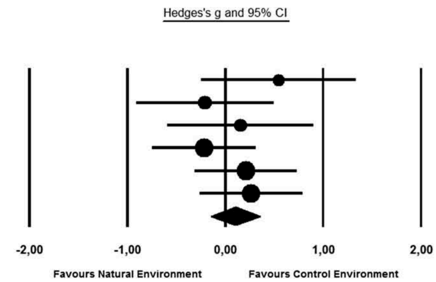

Figure 6. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to visual attention. No moderator variables are included.

Table 8. Summary of results from the visual attention meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Visual Attention					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	6	0.100 (−0.147 – 0.347)	0	0.793	0.427	
Moderators						
Baseline balance						Q = 0.031, DF = 1, p = 0.859
yes	3	0.083 (−0.218 – 0.386)	0	0.546	0.585	
partial	3	0.132(−0.296 – 0.559)	0	0.602	0.547	
no	-	-	-	-	-	-
unsure	-	-	-	-	-	-

Results of the meta-analysis pooled across all levels of the moderator variables (Figure 8 and Table 12) show that natural stimuli were found to exert no marked effect on impulse control, (n = 5). As illustrated in Figure 8B (supplementary material), there was also no effect of the environment when studies were analyzed based upon full (n = 3), or partial (n = 2), baseline balance. Due to a lack of sufficient data across more than two levels of restoration potential exposure type, further moderator analyses were not performed.

Processing speed

A total of 8 outcome measures derived from 7 cognitive tasks were categorized as being relevant to processing speed (Table 13). Outcome measures were categorized in this cognitive domain if they involved timed performance on tasks that were relatively short in duration, that is, not tasks measuring vigilance or sustained attention. The speed by which the tasks were performed may depend on mechanisms closely related to cognitive processing (e.g., Stroop Task, congruent trials) or occur somewhere in conjunction of cognitive and motor processing (e.g., substitution tasks). Substitution tasks require participants to fill-in as many letters or digits as possible in a given time period, based on a letter-digit- or symbol-key provided. There were 4 versions of these tasks used in the meta-analysis.

Results of the meta-analysis pooled across all levels of the moderator variables (Table 14 and Figure 9) demonstrate that natural environments or stimuli exerted no effect on processing speed, (n = 10). As observed in Figure 9B (supplementary material), there was also no significant effect of the environment when studies were analyzed based on full (n = 2), partial (n = 5), or no (n = 3) baseline balance. As Figure 9C and 9D (supplementary material) illustrate, there were no significant effects found when analyzing studies based on restoration potential or exposure type.

Cognitive flexibility

Three outcome measures derived from two cognitive tasks were categorized as being relevant to processing speed (Table 15). Outcome measures were included in this domain if performance involved switching between two cognitive sets, or sets of rules, within a task. This ability is sometimes called set-shifting and is classically assessed using the Wisconsin Card Sorting Task (Milner 1963). The Trail Making Task, version B,

Table 9. Data and moderator details from included studies using outcome measures classified in the vigilance domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
ANT (alerting score)	Berman, Jonides, and Kaplan (2008) (2)	-	12	32 (23.76)	31 (18.12)	12	36 (22.59)	46 (38.18)	none	-	T2†	partial	34.00	virtual
	Gamble, Howard, and Howard (2014)	older adults	13	25.81 (55.46)	45.87 (38.32)	13	49.94 (40.36)	49.17 (34.99)	none	-	T2†	partial	37.88	virtual
		younger adults	15	34.97 (26.26)	34.08 (32.60)	15	8.79 (48.16)	15.25 (30.47)	none	-	T2†	partial	21.88	virtual
SART (d-prime)	Berto (2005) (1)	-	16	1.40 (0.71)	1.86 (0.89)	16	1.97 (0.96)	2.00 (0.95)	none	-	T2†	partial	1.69	virtual
	Berto (2005) (3)	-	16	2.12 (1.08)	2.47 (1.04)	16	1.79 (1.24)	2.03 (0.93)	none	-	T2†	partial	1.96	virtual
SART (RT)	Berto (2005) (1)	-	16	313.17 (38.36)	267.38 (73.78)	16	319.59 (70.98)	299.61 (41.43)	none	-	T2†	partial	316.38	virtual
	Berto (2005) (3)	-	16	311.27 (35.6)	302.22 (32.09)	16	306.21 (78.79)	297.91 (52.98)	none	-	T2†	partial	308.74	virtual
SART (RT, first half)	Lee et al. (2015)	-	75	551.64 (149.91)	526.73 (159.35)	75	541.64 (139.69)	525.01 (172.34)	none	-	T2†	yes	546.64	virtual
SART (RT, second half)	Lee et al. (2015)	-	75	526.35 (157.01)	519.29 (168.79)	75	525.92 (160.56)	501.56 (158.05)	none	-	T2†	yes	526.14	virtual
SART (SD of RT, first half)	Lee et al. (2015)	-	75	97.30 (36.37)	90.67 (31.09)	75	95.37 (34.29)	102.27 (42.26)	none	-	T2†	yes	96.34	virtual
SART (SD of RT, second half)	Lee et al. (2015)	-	75	101.24 (43.73)	96.41 (35.07)	75	107.38 (41.14)	116.46 (49.10)	none	-	T2†	yes	104.31	virtual
SART (omission errors, first half)	Lee et al. (2015)	-	75	0.49 (1.30)	0.37 (0.95)	75	0.65 (2.60)	0.63 (1.65)	none	-	T2†	yes	0.57	virtual
SART (omission errors, second half)	Lee et al. (2015)	-	75	0.63 (1.73)	0.68 (1.39)	75	0.50 (1.13)	1.43 (1.47)	none	-	T2†	yes	0.57	virtual
MSIT (% error, congruent)	Jung et al. (2017)	healthy adults	20	0.36 (1.60)	0.00 (0.00)	20	0.05 (0.24)	0.05 (0.21)	none	-	T2†	partial	0.21	virtual
		heart failure patients	20	0.56 (1.18)	1.24 (4.42)	20	0.48 (0.88)	1.08 (1.96)	participants	-	T2*	partial	0.52	virtual
MSIT (RT, congruent)	Jung et al. (2017)	healthy adults	20	661 (93)	629 (79)	20	662 (122)	636 (127)	none	-	T2†	partial	661.50	virtual
		heart failure patients	20	755 (143)	737 (123)	20	741 (116)	721 (114)	participants	-	T2*	partial	748.00	virtual

Notes. ANT = Attention Network Task; SART = Sustained Attention to Response Task; MSIT = Multi-Source Interference Task. *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement

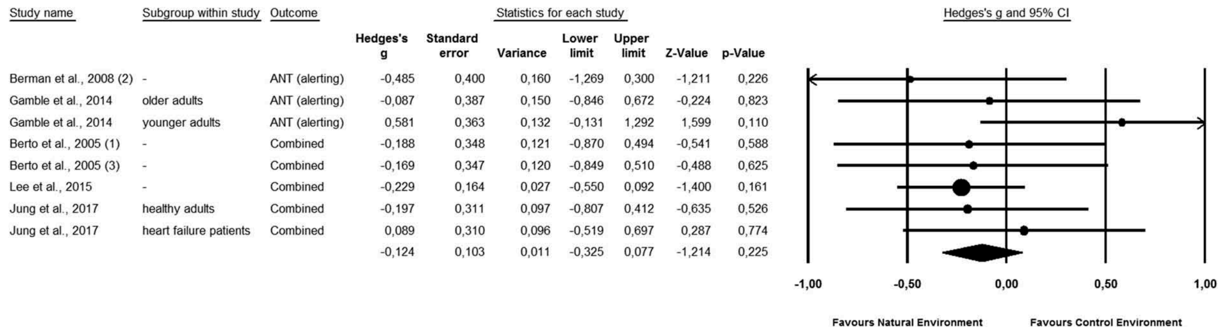

Figure 7. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to vigilance. No moderator variables are included.

Table 10. Summary of results from the vigilance meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Vigilance					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	8	−0.124 (−0.325⊠ 0.077)	0	−1,214	0.225	-
Moderators			Insufficient data			

involves switching between letter- or number-based cognitive sets, whereas the Switching Stroop Task, requires participants to switch between words and colors.

As shown in Figure 10 and Table 16, natural environments or stimuli were found to produce a significantly positive effect on cognitive flexibility when effect sizes were pooled across all levels of moderators (n = 6), with a pooled estimated mean effect size (Hedges' g) of −0.317. As Figure 10B (supplementary material) shows, the positive effect remains (Hedges' g = −0.610) when only including studies with reported baseline balance (n = 2). Figure 10C (supplementary material) demonstrates no significant effect found when only including participants with heightened restoration; however, significance was reached for studies with participants of normal restoration potential, (n = 3). Studies that used actual exposure to real environments reported significant positive effects (Hedges' g = −0.417) in favor of natural environments (n = 4), as seen in Figure 10D (supplementary material). However, the subgroup difference between actual and virtual exposure types did not reach significance.

Emerging cognitive domains

Three cognitive tasks that fulfilled the inclusion criteria for the meta-analysis were only used once. Therefore, it was not possible to conduct meta-analyses using their outcome measures. Instead, these represent potential new directions in attention restoration research where further studies need to be conducted before assessing their suitability. Table 17 summarizes the basic statistics and moderator characteristics of the three tasks.

The Delay of Gratification Task (Jenkin et al. 2017) measures the ability to effortfully control thoughts and behavior regarding a temporally salient reward in order to receive a greater reward at a later time. It is classified within tasks related to self-regulation and was shown to predict positive outcomes later in life (Mischel, Shoda, and Rodriguez 1989). Jenkin and colleagues (2017) reported a significant improvement in this ability after viewing videos of natural environments when compared to video of urban environments containing little traffic and few people (study 1), but not videos of urban environments with higher traffic flow and more people to create more noise (study 2).

Taylor's Aggression Paradigm was used by Wang and colleagues (2017) to measure provoked and unprovoked aggression after watching videos of natural and urban environments. Using ego-depletion (or task-induced cognitive fatigue) as a between-subjects factor, they found that participants with heightened restoration potential displayed lower levels of provoked aggression after watching natural

Table 11. Data and moderator details from included studies using outcome measures classified in the impulse control domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
SART (commission errors[a])	Berto (2005) (1)	-	16	1.81 (3.83)	2.06 (4.79)	16	3.25 (6.22)	1.62 (4.96)	none	-	T2†	partial	2.53	virtual
	Berto (2005) (3)	-	16	1.62 (2.5)	0.81 (1.42)	16	1.68 (2.62)	0.75 (1.52)	none	-	T2†	partial	1.65	virtual
SART (commission errors[a], first half)	Lee et al. (2015)	-	75	1.97 (2.17)	2.76 (2.68)	75	2.54 (2.68)	3.42 (3.20)	none	-	T2†	yes	2.26	virtual
SART (commision errors[a], second half)	Lee et al. (2015)	-	75	2.52 (2.42)	2.83 (2.68)	75	2.86 (2.60)	3.32 (2.68)	none	-	T2†	yes	2.69	virtual
SART (total correct response to target)	Berto (2005) (1)	-	16	11.68 (5.28)	13.62 (5.37)	16	13.25 (5.09)	13.00 (5.4)	none	-	T2†	partial	12.47	virtual
	Berto (2005) (3)	-	16	14.81 (5.55)	17.12 (4.09)	16	12.50 (6.36)	14.56 (5.95)	none	-	T2†	partial	13.66	virtual
SART (% total correct response to target)	Craig et al. (2015)	High depression symptom rating	12	63.19 (26.76)	63.54 (29.52)	12	66.32 (20.06)	62.85 (26.32)	participant	-	T2*	yes	64.76	virtual
		Typical depression symptom rating	12	63.54 (13.19)	76.74 (18.76)	12	64.58 (17.81)	47.57 (17.27)	none	-	T2*	yes	64.06	virtual

Notes. SART = Sustained Attention to Response Task. *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

[a]Incorrect response to target (unsuccessfully withheld response).

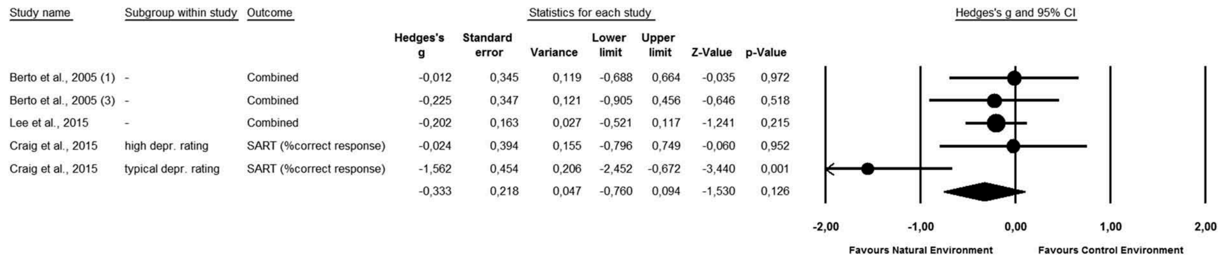

Figure 8. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to impulse control. No moderator variables are included.

Table 12. Summary of results from the impulse control meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Impulse Control					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	5	0.333 (−0.760 – 0.094)	56.68	−1.530	0.216	
Moderators						
Baseline balance						Q = 0.778, DF = 1, p = 0.378
yes	3	−0.528 (−1.302 – 0.247)	76.65	−1.335	0.182	
partial	2	−0.118 (−0.579 – 0.362)	0	−0.480	0.631	
no	-	-	-	-	-	-
unsure	-	-	-	-	-	-

environments compared to urban environments. For participants with normal restoration potential, provoked aggression was higher after watching natural environments. No significant effects were found for the unprovoked aggression measure.

Finally, Chow and Lau (2015) used the Graduate Record Exam (GRE) to measure logical reasoning abilities before and after viewing digital photographs of natural and urban scenes. The GRE is used to screen applicants for graduate studies in the United States and requires responders to apply mental models, make inferences, and draw logical conclusions to solve the problems (Schmeichel, Vohs, and Baumeister 2003). By analyzing change scores, Chow and Lau (2015) reported a significant positive effect on logical reasoning after viewing natural scenes compared to urban scenes.

Discussion

Key findings

Our systematic search of literature published after July 2013 uncovered 42 articles where the restorative effect of natural environments was investigated using objective outcome measures. The aim was to identify cognitive domains that are sensitive to this effect and consider features of associated outcome measures to better describe processes related to directed attention, the resource purported to be restored. Our search revealed a substantial increase in interest and productivity within the area. This suggests an urgency to address the ambiguity surrounding directed attention as the cognitive resource underlying the restoration effect, and the diversity of protocols used to measure it. A meta-analysis was conducted on all outcome measures from studies identified by ourselves and Ohly and colleagues (2016) that included baseline measurement. When analyzing across all levels of baseline balance, improved performance was found for the domains of working memory, attentional control, and cognitive flexibility, with low to moderate effect sizes. However, improvements in attentional control were not detected when only examining studies with fully balanced baseline measures. Moderator analyses revealed that actual exposures to real environments are likely to enhance the restoration effect within these three domains. However, it was typical for actual exposures to be longer than virtual exposures, suggesting exposure length might also explain the result. The effect of restoration potential among participants was less clear. These findings suggest

Table 13. Data and moderator details from included studies using outcome measures classified in the processing speed domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
DLST (total correct, 90s)	Van den Berg et al., 2017	2-month follow up	84	31.68 (8.99)	37.37 (9.39)	86	32.13 (6.67)	37.94 (8.47)	none	-	T2†	yes	31.91	actual
		4-month follow up	84	31.68 (8.99)	40.68 (10.09)	86	32.13 (6.67)	40.52 (8.62)	none	-	T2†	yes	31.91	actual
LDST (total correct, 60s)	Lymeus et al., 2017	prior to mindfulness training	9	42.00 (6.75)	41.89 (5.56)	7	42.43 (2.99)	44.71 (2.63)	participant		T2*	yes	42.22	virtual
		2 weeks mindfulness training	9	42.33 (7.53)	42.56 (4.50)	7	42.43 (5.44)	45.14 (3.81)	participant	-	T2*	yes	42.38	virtual
		4 weeks mindfulness training	9	43.44 (8.28)	43.00 (6.14)	7	43.14 (1.35)	41.29 (7.16)	participant	-	T2*	yes	43.29	virtual
		6 weeks mindfulness training	9	45.67 (9.78)	43.56 (7.78)	7	46.14 (4.53)	46.14 (4.18)	participant	-	T2*	yes	45.91	virtual
		8 weeks mindfulness training	9	43.78 (7.48)	45.44 (7.86)	7	45.71 (4.72)	44.43 (5.22)	participant	-	T2*	yes	44.75	virtual
SDMT (total correct, 90s)	Bodin & Hartig, 2003	men	6	51.17 (7.17)	48.83 (4.22)	6	51.50 (3.67)	49.17 (5.38)	none	-	T2†	partial	51.34	actual
		women	6	56.33 (11.33)	52.50 (10.04)	6	56.00 (9.36)	52.83 (9.56)	none	-	T2†	partial	56.17	actual
SST (total correct, 60s)	Johansson, Hartig, and Staats. 2011	alone	20	38.65 (5.28)	37.85 (5.10)	20	37.85 (5.21)	37.80 (4.87)	none	-	T2†	no	38.25	actual
		with friend	20	40.00 (6.78)	36.35 (5.09)	20	37.70 (4.78)	36.85 (4.79)	none	-	T2†	no	38.85	actual
Sky Search Task (total correct)	Van den Berg et al., 2016	2-month follow up	84	9.64 (3.18)	10.73 (3.33)	86	9.19 (3.04)	10.28 (3.48)	none	-	T2†	yes	9.42	actual
		4-month follow up	84	9.64 (3.18)	13.05 (3.10)	86	9.19 (3.04)	12.03 (3.11)	none	-	T2†	yes	9.42	actual
TMTA (completion time)	Cimprich & Ronis, 2003	-	83	30.23 (11.84)	25.65 (9.38)	74	37.08 (19.79)	31.21 (11.53)	participants	-	T2*	no	33.66	actual
	Jung et al., 2017	healthy adults	20	27.44 (9.67)	24.68 (6.92)	20	25.03 (7.28)	24.95 (7.57)	none	-	T2†	partial	26.24	virtual
		heart failure patients	20	41.90 (14.87)	41.80 (21.67)	20	41.92 (17.99)	44.38 (28.18)	participants		T2*	partial	41.91	virtual

(*Continued*)

Table 13. (Continued).

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
	Stark, 2003	-	29	22.06 (7.09)	19.84 (4.79)	28	21.67 (5.46)	19.60 (5.33)	participants	-	T2*	partial	21.87	actual
Stroop Task (% error, congruent)	Jung et al., 2017	healthy adults	20	0.29 (0.88)	1.17 (2.67)	20	1.17 (3.38)	0.44 (1.04)	none	-	T2†	partial	0.73	virtual
		heart failure patients	20	4.17 (8.14)	2.5 (3.24)	20	3.47 (4.84)	2.92 (3.55)	participants	-	T2*	partial	3.82	virtual
Stroop Task (RT, congruent)	Jung et al., 2017	healthy adults	20	1282 (300)	1178 (275)	20	1242 (332)	1180 (352)	none	-	T2†	partial	1262.00	virtual
		heart failure patients	20	1432 (476)	1352 (338)	20	1285 (304)	1281 (296)	participants	-	T2*	partial	1358.50	virtual
Stroop Task (% error, nonswitched)	Jung et al., 2017	healthy adults	20	8.89 (16.40)	4.01 (5.93)	20	6.76 (10.33)	4.54 (7.42)	none	-	T2†	partial	7.83	virtual
		heart failure patients	20	10.49 (15.69)	10.24 (14.28)	20	11.12 (17.21)	8.66 (12.78)	participants	-	T2*	partial	10.81	virtual
Stroop Task (RT, nonswitched)	Jung et al., 2017	healthy adults	20	1578 (479)	1333 (540)	20	1408 (441)	1341 (484)	none	-	T2†	partial	1493.00	virtual
		heart failure patients	20	1485 (405)	1370 (416)	20	1509 (519)	1492 (447)	participants	-	T2*	partial	1497.00	virtual

Notes. DLST = Digit-Letter Substitution Task; LDST = Letter-Digit Substitution Task; SDMT = Symbol-Digit Modalities Task; TMTA = Trail Making Task A *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

Table 14. Summary of results from the processing speed meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Processing Speed					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	10	−0.138 (−0.299 – 0.022)	0	−1.687	0.092	-
Moderators						
Baseline balance						Q = 0.997, DF = 2, p = 0.607
yes	2	−0.072 (−0.357 – 0.214)	0	−0.490	0.624	
partial	5	0.049 (−0.252 – 0.350)	0	0.322	0.748	
no	3	−0.215 (−0.648 – 0.218)	55.59	−0.972	0.331	
unsure	-	-	-	-		
Restoration potential						Q = 0.369, DF = 1, p = 0.544
heightened	4	−0.161 (−0.541 – 0.219)	49.32	−0.832	0.406	
normal	6	−0.025 (−0.243 – 0.192)	0	−0.229	0.819	
Exposure type						Q = 1.200, DF = 1, p = 0.273
actual	7	0.163 (−0.368 – 0.042)	16.05	−1.557	0.119	
virtual	3	0.084 (−0.308 – 0.476)	0	0.422	0.673	

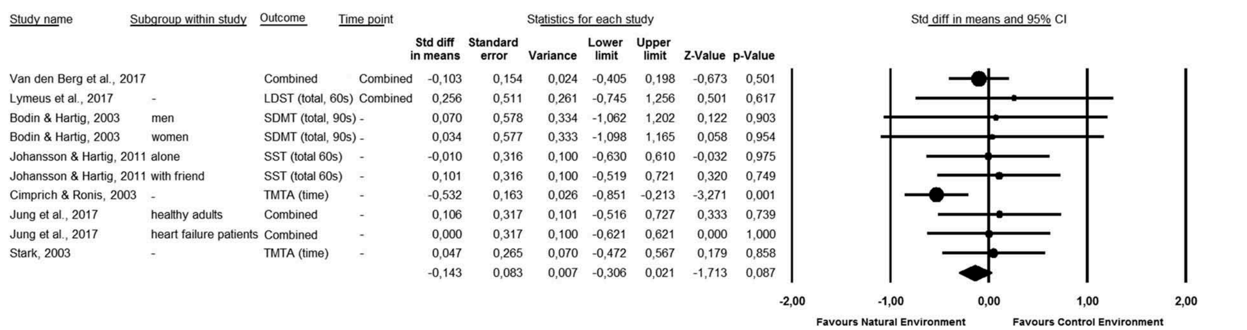

Figure 9. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to processing speed. No moderator variables are included.

the construct of directed attention, and restoration effect of exposure to natural environments needs to be updated using existing evidence to guide future research.

Conceptualizing directed attention in attention restoration theory

Kaplan and Kaplan (1989) adopted the term 'directed attention' to help distinguish from William James' (1892) concept of 'voluntary attention' from which it was inspired. This enabled the term to be used as a distinct construct under the scope of ART. In the original conceptualizations of directed attention within ART (Kaplan 1995; Kaplan and Berman 2010; Kaplan and Kaplan 1989), there appear to be three characteristics central to its nature.

I. Directed attention is a finite capacity that requires voluntary cognitive effort

James (1892) suggested the central component to voluntary attention was a cognitive effort. In order to obtain a desired goal or outcome, a degree of voluntary effort is required to focus on certain stimuli. The cognitive effort required to successfully perform a task can be determined by an interaction of perceptual, sensory, and cognitive loads (Sörqvist et al. 2016). Perceptual load is determined by the ratio of task-relevant to task-irrelevant stimuli present; sensory load is determined by the ease or difficulty of which task-relevant stimuli may be processed; and cognitive load is determined by the extent to which a task recruits cognitive resources. Cognitive effort is often conceptualized through these three factors in terms of resistance to distraction during task performance. For example, when a task has a high perceptual load (through a large number of task-relevant stimuli) or high sensory load (through more salient task-related stimuli), it is easier to

Table 15. Data and moderator variables from included studies using outcome measures classified in the cognitive flexibility domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
TMTB (completion time)	Cimprich and Ronis (2003)	-	83	65.01 (30.98)	56.51 (26.60)	74	77.09 (42.15)	68.01 (34.93)	participants	-	T2*	no	71.05	actual
	Jung et al. (2017)	healthy adults	20	58.10 (22.37)	53.03 (22.32)	20	56.88 (20.66)	55.25 (21.84)	none	-	T2†	partial	57.49	virtual
		heart failure patients	20	97.78 (47.52)	87.37 (45.35)	20	98.29 (52.80)	88.04 (44.39)	participants	-	T2*	partial	98.04	virtual
	Shin et al. (2011)	first walk	30	37.03 (6.81)	29.48 (6.82)	30	37.03 (6.81)	39.24 (21.23)	none	-	T2†	yes	37.03	actual
		second walk	30	37.04 (6.90)	29.45 (6.72)	30	37.04 (6.90)	39.17 (21.23)	none	-	T2†	yes	37.04	actual
	Stark (2003)	-	29	46.55 (11.92)	38.89 (10.75)	28	48.15 (10.42)	40.40 (9.30)	participants	-	T2*	partial	47.35	actual
Stroop Task (% error, switched)	Jung et al. (2017)	healthy participants	20	6.04 (7.13)	1.05 (3.15)	20	5.88 (10.56)	1.40 (4.35)	none	-	T2†	partial	5.96	virtual
		heart failure patients	20	12.75 (14.99)	7.16 (8.71)	20	9.63 (13.80)	6.55 (8.19)	participants	-	T2*	partial	11.19	virtual
Stroop Task (RT, switched)	Jung et al. (2017)	healthy participants	20	1547 (515)	1311 (410)	20	1515 (536)	1283 (417)	none	-	T2†	partial	1531.00	virtual
		heart failure patients	20	1619 (693)	1654 (547)	20	1373 (376)	1475 (464)	participants	-	T2*	partial	1496.00	virtual

Notes. TMTB = Trail Making Task B. *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

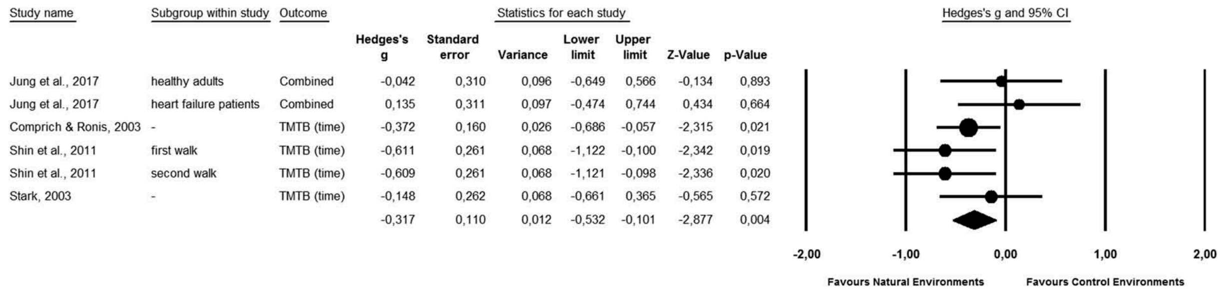

Figure 10. Overall forest plot showing pooled effect sizes (Hedges' *g*) for outcome measures related to cognitive flexibility. No moderator variables are included.

Table 16. Summary of results from the cognitive flexibility meta-analysis of post-test scores and moderator analyses exploring the influence of baseline balance, restoration potential, and exposure type.

	Cognitive Flexibility					
	n	Hedge's G (95% CI)	I^2	Z-value	p-value	Test of heterogeneity between subgroups
Overall	6	−0.317 ('-0.532 – '0.101)	16.04	−2.877	**0.004**	-
Moderators						
Baseline balance						Q = 5.470, DF = 2, p = 0.065
yes	2	−0.610 (−0.972 – −0.249)		−3.308	**0.001**	
partial	3	−0.034 ('-0.364 – 0.296)		−0.201	0.840	
no	1	−0.372 (−0.686 – −0.057)		−2.315	**0.021**	
unsure	-	-	-	-	-	
Restoration potential						Q = 1.055, DF = 1, p = 0.304
heightened	3	−0.224 (−0.493 – 0.044)	11.09	−1.639	0.101	
normal	3	−0.454 (−0.802 – −0.107)	19.44	−2.565	**0.010**	
Exposure type						Q = 3.560, DF = 1, p = 0.059
actual	4	−0.417 (−0.632 – −0.201)	0	−3.793	**0.000**	
virtual	2	0.046 (−0.384 – 0.477)	0	0.211	0.833	

ignore distracting task-irrelevant stimuli (Lavie et al. 2004). In this case, it is believed that when a task demands high cognitive effort due to perceptual or sensory processes, the necessary increase in attentional control facilitates the inhibition of distractors. Conversely, when a task demands high cognitive effort due to high cognitive load, the necessary increase in attentional control places higher demands on cognitive resources that are limited, which causes an attenuation of the ability to ignore distractors (Lavie 2005).

II. Directed attention requires focus on task-relevant stimuli and simultaneous inhibition of task-irrelevant distractors

Directed attention has been described loosely as a "*... means of achieving focus in a confusing environment*" (Kaplan and Kaplan 1989, p. 188) and more succinctly as the ability to "*fix attention on certain stimuli while suppressing other stimuli*" (Kaplan and Berman 2010, p. 44). These definitions highlight another central process required for directed attention, the inhibition of distraction in order to maintain focus (Kaplan 1995). This skill is sometimes referred to broadly as interference control, where distractors may be internal or external (Diamond 2013). Internal distractors, such as unwanted thoughts or memories, are ignored through the process of cognitive inhibition. Whereas inhibition of external distractors takes place at the perceptual level where target stimuli are attended to while others are ignored (Diamond 2013). In earlier definitions of directed attention in the ART literature, it was not explicit whether cognitive inhibition, or the suppression of internal distractors, was included. However, Kaplan and Berman (2010) suggested that this process might also be sensitive to restorative environments.

III. Directed attention is a resource recruited by other cognitive processes

Kaplan and Berman's (2010) extension of the directed attention construct adopted by ART appears to have done service to the field by bringing the theory to a wider readership with

Table 17. Data from included studies using outcome measures classified as emerging cognitive domain.

			Natural environment/stimuli			Control environment/stimuli			Moderators					Exposure
				Pre	Post		Pre	Post	Restoration potential		Baseline			Exposure
Cognitive task (outcome measure)	Author, Year, (sub-study)	sub-group	*n*	mean (SD)	mean (SD)	*n*	mean (SD)	mean (SD)	Source of RP	Induction length	Baseline time	Baseline balance	Baseline average	Exposure type
DoG (total delayed choices)	Jenkin et al. (2017) (1)	-	25	3.12 (1.51)	3.44 (1.36)	25	3.20 (1.87)	2.52 (1.61)	none	-	T2†	yes	3.16	virtual
	Jenkin et al. (2017) (2)	-	26	3.04 (1.78)	3.27 (1.25)	28	3.46 (1.04)	2.96 (1.07)	none	-	T2†	yes	3.25	virtual
TAP (unprovoked)	Wang et al. (2017)	non-depletion	29	0.05 (0.89)	0.02 (0.86)	29	−0.02 (0.74)	−0.09 (0.87)	none	-	T2†	yes	0.02	virtual
		depletion	31	−0.17 (0.78)	−0.11 (0.79)	29	0.14 (0.82)	0.20 (0.99)	task	NR	T1	yes	−0.02	virtual
TAP (provoked)	Wang et al. (2017)	non-depletion	29	−0.17 (0.87)	−0.07 (0.75)	29	−0.07 (0.77)	−0.23 (0.86)	none	-	T2†	yes	−0.12	virtual
		depletion	31	−0.05 (0.85)	−0.08 (0.91)	29	0.30 (0.95)	0.38 (0.83)	task	NR	T1	yes	0.13	virtual
GRE (total correct)	Chow and Lau (2015) (2)	-	29	6.00 (2.51)	7.31 (2.49)	29	7.28 (3.07)	7.21 (2.47)	task	6″00′	T1	unsure	6.64	virtual

Notes. DoG = Delay of Gratification Task; TAP = Taylor's Aggression Paradigm; GRE = Graduate Record Exam. *Participants with heightened restoration potential at baseline measurement; †Participants without heightened restoration potential at baseline measurement.

the inclusion of executive functioning and self-regulation. The extension is perhaps responsible for the diversity in outcome measures identified. Kaplan and Berman (2010) suggested that other cognitive processes required to perform tasks associated with these outcome measures may also be sensitive to the restoration effect. Indeed, models exploring the interrelatedness of executive functions do not necessarily place primacy on directed attention, but rather considers it as one type of inhibitory control that interacts with other executive abilities to subserve goal-directed behavior (Diamond 2013).

The involvement of directed attention in cognitive tasks cannot be quantified; however, it is possible to examine aspects of these tasks to ascertain a better description of directed attention, as a cognitive resource sensitive to the restoration effect. Stimuli contained in cognitive tasks can be categorized into those to be attended (targets) and those to be suppressed (distractors). This is obvious in certain tasks measuring the ability to suppress external distraction, such as the Flanker Task (Eriksen 1995) where the center arrow acts as the target to be attended to while the flanking arrows act as distractors. However, Kaplan and Berman (2010) explained that tasks measuring aspects of self-regulation rely on a comparable ability of suppressing internal distractors. An example of this was demonstrated by Baumeister et al. (1998) findings on self-control where participants forced themselves to eat radishes rather than chocolate by overriding a preferred response and suppressing internal cues associated with the more favored food choice. The investigators manipulated the situation by asking participants to skip one meal before the session and by having the experimental area smell of fresh chocolate and baking. Thus, the demand for suppression of internal cues (hunger) was heightened through the senses. Those who exerted the necessary self-control were less likely to persist on a subsequent unsolvable puzzle task, suggesting that suppression of internal (and external) distraction depleted a cognitive resource under a process they called ego-depletion (Baumeister et al. 1998). Indeed, several studies in our review referred to fatigue induction as ego-depletion (Chow and Lau 2015).

Expanding the definition of directed attention

Based upon the characteristics of directed attention described above, a framework is proposed in which to categorize the cognitive domains and outcome measures found to be sensitive to the restorative effect of natural environments. It is conceivable that the framework provides a useful base from which to unpack our findings so that directed attention can be better described based upon current evidence.

The Three D’s of Directed Attention

It is proposed that outcome measures used within ART research may be categorized along three axes or continuums: 1) cognitive *Demand* (high–low), based on levels of perceptual, sensory, and cognitive loads; 2) *Direction* of attentional focus (internal–external), whether goal-relevant task components are stimuli that are perceived externally or information that is represented internally; and 3) locus of *Distraction* (internal–external), whether goal-irrelevant task components are stimuli that are perceived externally or information that is represented internally. Each axis is described in more detail below and then illustrated by using representative examples from the three cognitive domains found to be positively influenced by natural environments during our meta-analysis.

1) Cognitive Demand

The ability to direct attention requires voluntary cognitive effort. Tasks purported to measure this ability vary in the amount of effort that is required to perform them successfully. This variation may occur between task types but also within different manipulations of the same task. As described earlier, the cognitive effort required to perform a task is determined by the associated perceptual, sensory, and cognitive loads (Lavie et al. 2004). For example, Chow and Lau (2015) enhanced cognitive demand through the high sensory load by increasing the salience of distracting thoughts using hunger and smell. Loadings may be combined to give an overall cognitive demand rating. Tasks with the highest cognitive demand, deemed most difficult to perform, are predicted to have: low perceptual load of targets, where the number of task-relevant stimuli is lower than the number of task-irrelevant stimuli; low sensory load, where the salience of task-related stimuli is low; and high cognitive load, where task performance requires the simultaneous use of other cognitive resources. Conversely, tasks with the lowest cognitive demand, deemed easiest to perform, are predicted to have: high perceptual load, where the ratio of task-relevant to task-irrelevant stimuli is high; high sensory load, where task-relevant stimuli are salient; and low cognitive load, where the task is performed using directed attention unencumbered by other cognitive capacities.

2) Direction of Attentional Focus

This component of directed attention tasks deals with the target stimuli. Target stimuli are identified as those to which attention needs to be directed in order to successfully perform the task. Following an earlier taxonomy of attention (Chun, Golomb, and Turk-Browne 2011), target stimuli were categorized as either external or internal. External stimuli are those that are processed using sensory modalities – vision, hearing, touch, smell, and taste. Chun and colleagues (2011) explained that attention may be directed towards each separately or in combination in order to achieve a required goal. Attention may also be directed towards spatial and temporal characteristics of the target stimuli, features of the stimuli or the stimuli as complete objects (Chun, Golomb, and Turk-Browne 2011). Internal target stimuli refer to goal-related representations that exist internally among goal-unrelated representations held in working memory and long-term memory, which may relate to task rules, decisions, and responses (Chun, Golomb, and Turk-Browne 2011).

3) Locus of Distraction

The locus of distraction refers to the way in which goal-unrelated stimuli are presented and processed during task performance. While target stimuli may be categorized as being identified through sensory modalities or selected from internal representations (Chun, Golomb, and Turk-Browne 2011), distractors need to be successfully ignored as part of the same identification and selection processes. Therefore, tasks measuring directed attention may be associated with either or a combination of external and internal distractors that are manipulated purposefully by the tasks' demands. The use of a continuum, rather than categorical ratings, seems most appropriate as there is likely to be an interplay between how external distractors influence internal distractors, and vice versa.

Figure 11 presents an example of hypothetical ratings using the 3Ds of Directed Attention framework with three different outcome measures. For the executive score of the ANT from the attentional control domain: Cognitive demand is relatively high for this task due to moderately low target to distractor ratio (perceptual load), non-salient stimuli (sensory load), and added conflict processing from congruent versus incongruent trials (cognitive load). Other versions of the task might heighten cognitive demand by including more distractor stimuli when the target is displayed. The direction of the attentional focus is purely external as the target stimuli are present only on the screen. Similarly, the locus of distraction is purely external as distractor stimuli are present only on the screen.

For the DSB Task from the working memory domain: Cognitive demand is high for this task due to high cognitive load where non-salient target stimuli must be integrated into the internal mental workspace (Hasher and Zacks 1988) as well as manipulated with other mental representations of target stimuli to be remembered. The direction of attentional focus is balanced between internal and external. In this task, goal-relevant stimuli are perceived externally, for example, audially, and then needs to be held as internal representations. The locus of distraction is purely internal as distractors include only thoughts, memories, and representations needing to be suppressed to keep the mental workspace clear. All sensory stimuli are goal-related in this task.

For the TMTB from the cognitive flexibility domain: Cognitive demand is high because of the need to switch between competing rule sets while also holding internal representations of them in order to do so. Further, there is always only one non-salient target stimulus in the array which changes after each is located. The direction of the

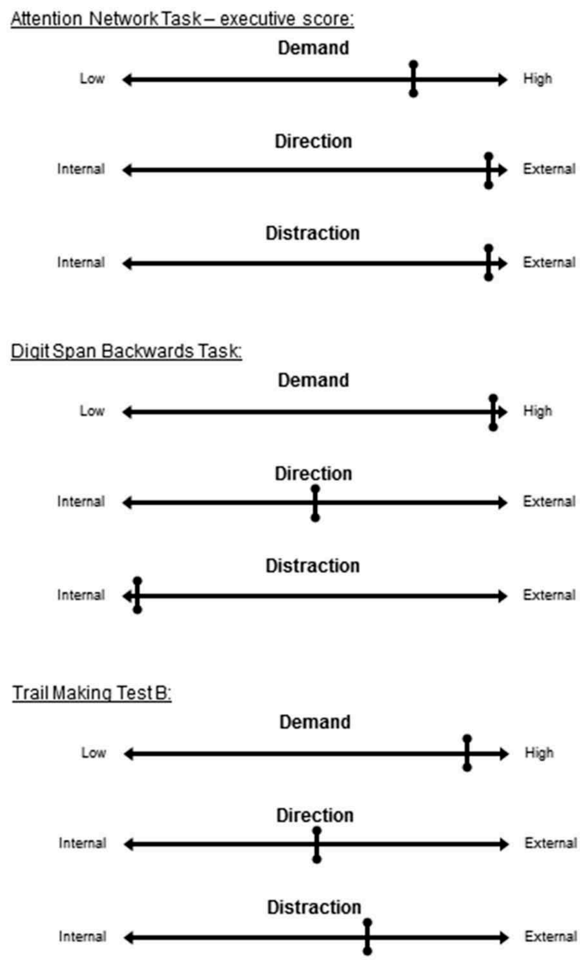

Figure 11. Hypothetical application of the 3Ds of Directed Attention framework using examples from the review. Using the three axes (demand, direction, distraction) to describe and compare outcome measures used in attention restoration studies allows researchers to consider in more detail the processes being influenced by the environment under their experimental conditions.

attentional focus is both external and internal. Goal-relevant stimuli are found visually, while internal representations of rule sets must also be attended to in order to perform the task. The locus of distraction is primarily external where all but one visual stimulus are distractors. However, it is also possible that perceiving these goal-irrelevant stimuli may evoke internal representations that are needed to be suppressed in order to successfully attend to the goal-relevant representation.

Study designs and future research

By including moderator analyses during our meta-analyses it was possible to explore whether variations in study design affected the effect sizes of pooled standardized means. The inclusion of baseline measures was of primary importance; however, there were limitations in how this information might be utilized. It is recommended that future studies using group randomization adjust for baseline using ANCOVA, while studies where group allocation is not randomized depend on assumptions of expected group differences (Van Breukelen 2013).

Our analysis revealed that actual exposure to natural environments significantly influenced effect sizes of working memory, attentional control, and cognitive flexibility, compared to virtual exposures. ART predicts that actual experiences in nature will restore cognitive abilities through inducing a psychological state of restoration. If virtual nature is sufficient to restore cognitive abilities, this suggests that a restorative state may be induced by visual (and sometimes auditory) representations alone and that these are sufficient. While this may be the case in several individual studies, it was found that exposure to actual restorative environments provides a stronger, more reliable effect and is therefore recommended for future researchers, where possible.

With that in mind, an alternative interpretation of the results, concerning the length of exposure, should also be addressed. Across the studies, it was typical for actual exposures to be longer than virtual exposures. Therefore, the moderating effect of exposure type may also be explained by exposure length. It is recommended future investigators consider both type *and* length of exposure when designing restoration studies.

Finally, heightened restoration potential, through fatigue induction or participant characteristics, strengthened the restoration effect on working memory, but not attentional control or cognitive flexibility. This finding goes directly against ART, in that heightened restoration potential should produce a more reliable restoration effect, if the mechanism underlying the effect is indeed restoration of a depleted cognitive resource. Essentially, this means there was mixed evidence for a restoration mechanism during our meta-analyses. If this observation was to be taken at face value, then one would need an additional explanation of how natural environments improve cognitive performance, which may act directly or indirectly with the restoration effect, such as the immune system boost hypothesized by Kuo (2015). However, it may also suggest that binary categorization of restoration potential is not appropriate and restoration potential is something that needs to be addressed at an individual level. For researchers, it should be encouraged to include an assessment of the success of fatigue induction as insufficiently fatigued participants may obscure results. A measure of fatigue, physiological or cognitive, before and after the fatigue induction period might serve as an optimal manipulation check. The time points presented in Figure 1 offers a suggestion of where these may be placed a restoration protocol.

Limitations

A key limitation for the current review was synthesizing post-test data from studies that varied widely in intervention, participant groups, and ART-related protocols. As also noted by Ohly and colleagues (2016), it highlights the need to synthesize a more homogeneous set of studies to ensure a more reliable synthesis. However, a more homogeneous set of experiments cannot occur without clarification of the directed attention construct and selection of appropriate outcome measures. It is hoped our review and proposed framework might help guide future studies.

The inclusion of baseline measurement was of primary importance to our synthesis; however, dealing with baseline in meta-analysis is often not straightforward. Testing for baseline differences within a study is considered misleading and is not recommended (Moher et al. 2012). Further, it is often not possible to use the preferred method of adjustment using ANCOVA during meta-analysis, unless all participant-level data are available. For study-level analysis, it is recommended to use only trials with good to moderate baseline balance to calculate treatment effects (Riley et al. 2013). If significant effects based on post-test data are lost, then one might conclude that individual-level data are needed to adequately adjust for baseline. This was the case for the attentional control cognitive domain, suggesting any conclusions drawn from post-test-only data for this domain should be considered with caution.

It is also difficult to apply the stringent measures of study quality relevant in other fields to the context of attention restoration research. One obvious problem is that environmental interventions cannot be blinded. It is therefore responsible to acknowledge the potential for something akin to a placebo or expectation effect when discussing how being in a natural environment influences brain function. This is especially true when one considers that most studies uncovered in the literature search used adult populations from developed countries where there likely exists cultural narratives around the idea that contact with nature is good for human health. Whether cognitive restoration from exposure to natural environments is a phenomenon relevant only to societies with such a cultural narrative is yet to be explored.

Conclusions

Our updated literature review and meta-analysis showed that working memory, cognitive flexibility and to a less-reliable degree, attentional control, may be improved after exposure to nature through restoration of cognitive processes related to directing attention. When considering the role of directed attention within these cognitive domains, it is not possible to quantify how much is recruited. However, one can qualify how directed attention is recruited during tasks within these cognitive domains. It is suggested that the Three D's of Directed Attention framework may be useful for this purpose. Within the framework, it is proposed cognitive domains and their related outcome measures may vary along axes of the *Demand* imposed by simultaneous cognitive processes; *Direction* of attentional focus towards targets or goal-relevant stimuli; and locus of *Distraction*, produced by distractors or goal-irrelevant stimuli. When observing the current evidence within this framework, data indicate that cognitive processes found to be sensitive to the restoration effect of natural environments are: 1) more likely to be high in cognitive demand, requiring the maintenance of several processes simultaneously; 2) related to the control of attention towards both internal and external goal-relevant stimuli; and 3) related to the suppression of internal distractors, such as goal-irrelevant thoughts, as well as external distractors, such as goal-irrelevant stimuli. It is hoped this information may guide future studies that are needed to better understand the actual process of attention restoration and how it relates to an individual's need for restoration or restoration potential.

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